

HAND 2 HAND

**Application Connecting Unskilled
Labour With Customers Developed Using MERN Stack**

PRESENTED BY:

Hema Sri Satya Gowri Narala (S200626)

V.D.N.V.Harika (S200356)

Ch.Kaveri (S200739)

G. Pavithra S200975)

Shaik Suneha (S200831)

B.Tejaswi (S200228)



**UNDER THE GUIDENCE OF Dr. H SRINIVASA
VARAPRASAD, M.Tech,Ph.D**

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Problem Statement

Unskilled daily-wageworkers often struggle to find regular work because they lack a digital presence, while customers find it difficult to locate reliable local laborers. Existing platforms mainly focus on skilled services, leaving a gap for unskilled workers who need better visibility and fair opportunities. There is a need for a simple and inclusive digital platform that helps unskilled workers create profiles, connect with nearby customers, and receive fair payments — all in a transparent and trustworthy way.



ABSTRACT

Unskilled daily-wage laborers often lack digital visibility, leading to irregular employment opportunities, while customers struggle to find reliable local workers for basic services. A prototype platform developed to bridge this gap through a mobile-first system that connects laborers and customers. This project designs the platform using the MERN stack (MongoDB, Express.js, React, Node.js) and implementing a robust role-based architecture with a strong focus on a secure and scalable backend. The system supports two primary roles—workers and customers—with features such as worker profile management, service listings, search and filtering, transparent wage agreements, chat system within the application, ratings and reviews system and secure interactions. The platform also promotes inclusivity through simplified onboarding and portfolio-style worker profiles. By strengthening backend connectivity, role-based access control, and system reliability, the project aims to deliver a functional prototype that improves worker visibility, builds customer trust, and creates a more transparent and digitally inclusive ecosystem for daily-wage employment.

INTRODUCTION

The HAND2HANDAPP is a mobile-first platform designed to connect customers with local daily-wage workers and service providers. Many unskilled workers have limited digital visibility, making it difficult to find consistent employment, while customers often struggle to identify reliable workers for basic services. To address this gap, the platform enables direct, location-based connections between workers and customers through a simple and accessible interface.

This project designs the system using the MERN stack (MongoDB, Express.js, React.js, and Node.js) with a role-based architecture. Customers can search and hire workers, while workers can manage their profiles, list services, and showcase their experience. By focusing on a secure and scalable backend, simplified onboarding, and transparent interaction flows, the platform aims to improve worker visibility, build customer trust, and create a more inclusive digital ecosystem for daily-wage employment.

LITERATURE SURVEY

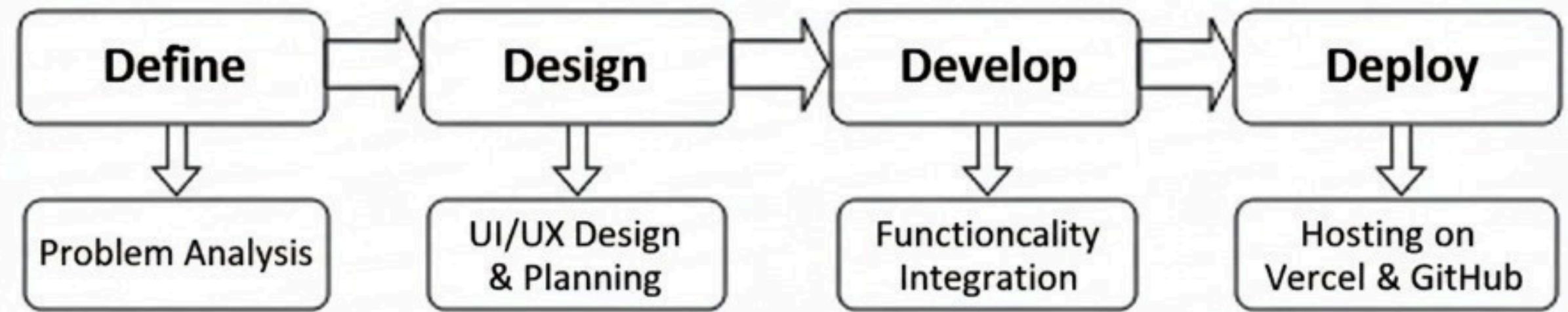
Theme / Area	Key Findings (with Authors)
Platformisation & Worker Precarity	Digital labour platforms increase convenience but often worsen precarity through commissions & opaque rules (M. Graham, Hjorth & Lehdonvirta [1]; Woodcock et al. [2]).
Urban Company (India)	Strong service-matching UX & quality monitoring. Worker protests highlight unfair rating policies, onboarding costs, and ID blocking (Anwar, Thomas, Toyama & Hui [3]; Dhar & Thuppilikkat [4]; BHRRC [5]).
HouseJoy & Local Apps	Popular in metros, offer convenient bookings. Exclude many unskilled daily-wage workers and charge high commissions (Woodcock et al. [2]).
JustDial / Sulekha	Wide reach through aggregator listings, useful for discovery. Weak verification & low trust in worker–customer ties (Woodcock et al. [2]).
Ola Tasker (pilot)	Leveraged Ola’s customer base. Failed due to trust & retention issues (Woodcock et al. [2]).
Informal Platforms (WhatsApp/Facebook groups)	Extremely low entry barrier, trusted in communities. No structured payment, dispute resolution, or reputation systems (Woodcock et al. [2]).
Reputation Systems	Ratings build trust but are biased; single averages oversimplify performance (HBR [6]; King’s College [7]).
Payments & Escrow	Escrow/staged payments reduce disputes, improve reliability (Woodcock et al. [2]; Medianama [8]).
Low-literacy Access	Voice/IVR systems enable digital adoption for low-literate workers (Patel et al. [9]; Tap2K [10]).

Gaps And Design Implementation

Identified Gaps	Design Implications
Under-representation of daily-wage workers (Graham et al. [1]).	Transparent wage terms and inclusion of unskilled categories (Woodcock et al. [2]; Medianama [8]).
Onboarding excludes low-literate workers (Patel et al. [9]; Tap2K [10]).	Voice-enabled onboarding with localized prompts + keypad confirmations.
Opaque rating & deactivation rules (Anwar et al. [3]; Dhar & Thuppilikkat [4]).	Multi-dimensional ratings (punctuality, workmanship, behaviour) instead of single average (HBR [6]; King's [7]).
Payment mistrust & disputes (Medianama [8]).	Lightweight escrow / advance payments to reduce cancellations & opportunism.
Weak worker credibility signals	Worker portfolios with images + audio appreciation/testimonials (Tap2K [10]; Garrett et al. [11]).

METHODOLOGY

“From concept to deployment: Our development plan.”



This project was developed following the 4D Model of Development, which includes the stages Define, Design, Develop, and Deploy. This structured model ensured an organized process Application — a platform designed to connect unskilled labourers with customers efficiently.

Technology Stack

MERN Stack (MongoDB, Express.js, React Native, Node.js)

- Unified JavaScript-based full stack enabling faster development and easier maintenance
Scalable and high-performance
- Backend using Node.js and Express.js for API development MongoDB for flexible and efficient data storage
- React Native for building cross-platform mobile applications with reusable components.

Technology Stack



SECURITY

- bcrypt – Password hashing
- jsonwebtoken (JWT) – Authentication

FILE HANDLING

- Cloudinary – Image storage
- Multer + Cloudinary Storage – File uploads

REAL TIME

- Socket.io – Live updates & communication

Main Feature And User Roles

Section	Main Activities	Roles
User Registration	Users (workers/customers) sign up and log in to access features.Worker, Customer	Worker, Customer
Worker Profile	Workers create profiles with skills, experience, and availability.Worker	Worker
Job Posting & Hiring	Customers post jobs; workers view and apply. Customers hire suitable workers. Customer, Worker	Customer, Worker
Job Listing View	Workers search for available jobs; customers browse nearby workers.Worker, Customer	Worker, Customer

Main Feature And User Roles

Section	Main Activities	Roles
Chat System	Allowing Customers and workers to communicate with eachother in the application itself	Customer, Worker
Interaction	Customers and workers communicate for job confirmation and details.	Worker,Customer
Search	Allowing Customers and workers to narrow down results based on conditions	Worker,customer
Ratings & Reviews	Customers provide ratings and reviews for workers after service completion to improve trust and service quality	Worker,Customer

Main Feature And User Roles

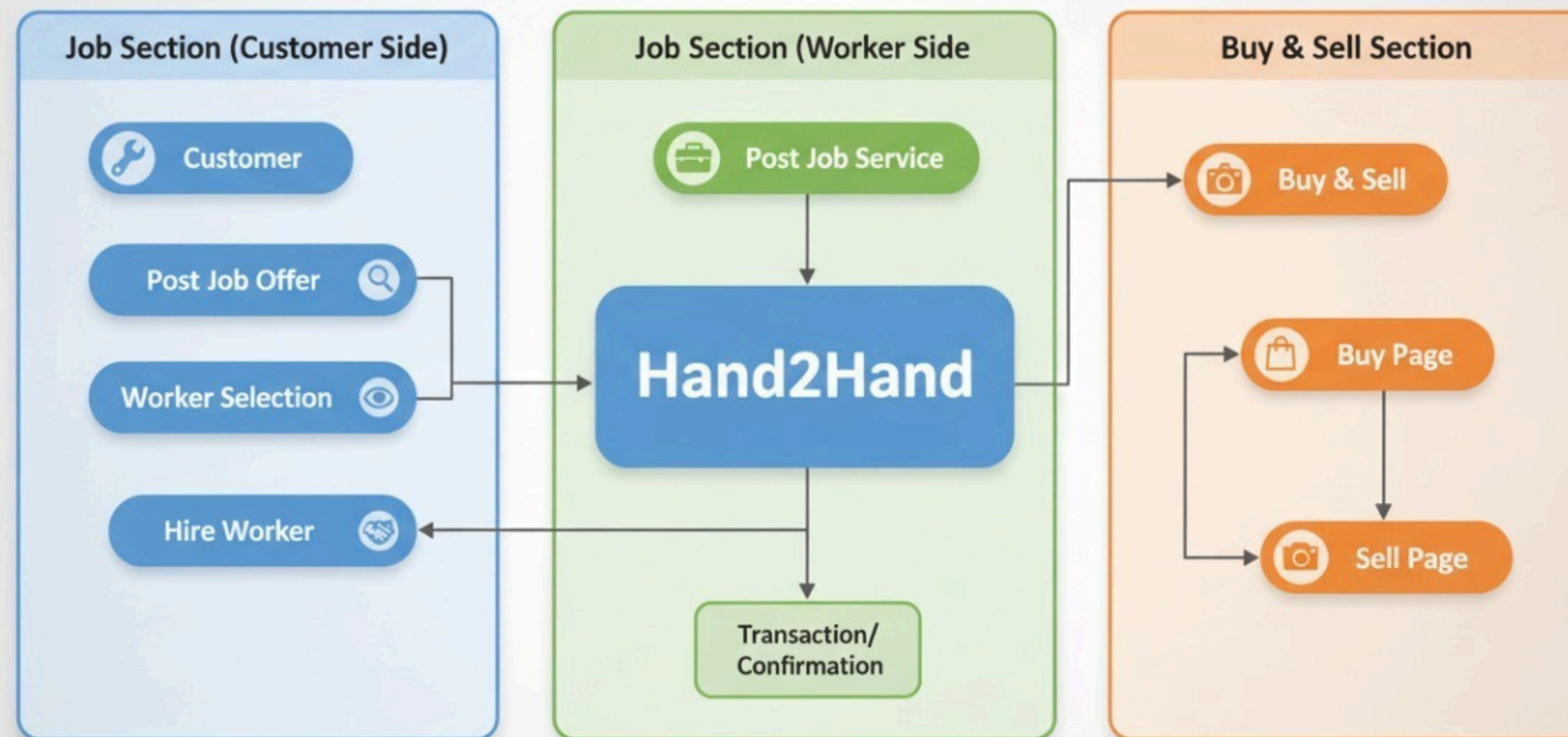
Section	Main Activities	Roles
Security Check	Asking security questions for user verification (used in forgot password)	Worker,Customer
Bookings Section	Managing bookings and displaying status (Active, Completed, Cancelled)	Worker,Customer
Filters	Allowing users to filter results based on conditions (price, category, etc.)	Worker,customer
Statistics	Showing user stats like Jobs Booked, Active Jobs, Saved Workers, and Total Spent	Worker,Customer

Identified Problems and Proposed Solutions

S.No	Problem	Benefit
1	Lack of trust between customers and workers	The system includes *ratings and reviews* to build trust. Customers can rate workers after completion of work, helping others choose reliable workers.
2	Unreliable or misbehaving workers	Ratings and reviews to the workers helps in encouraging better service to the customers
3	Sharing personal mobile numbers between customers and workers can lead to privacy issues and unwanted communication.	The platform includes a chat system that allows customers and workers to communicate safely without sharing personal contact details if they don't want to,improving privacy and safety

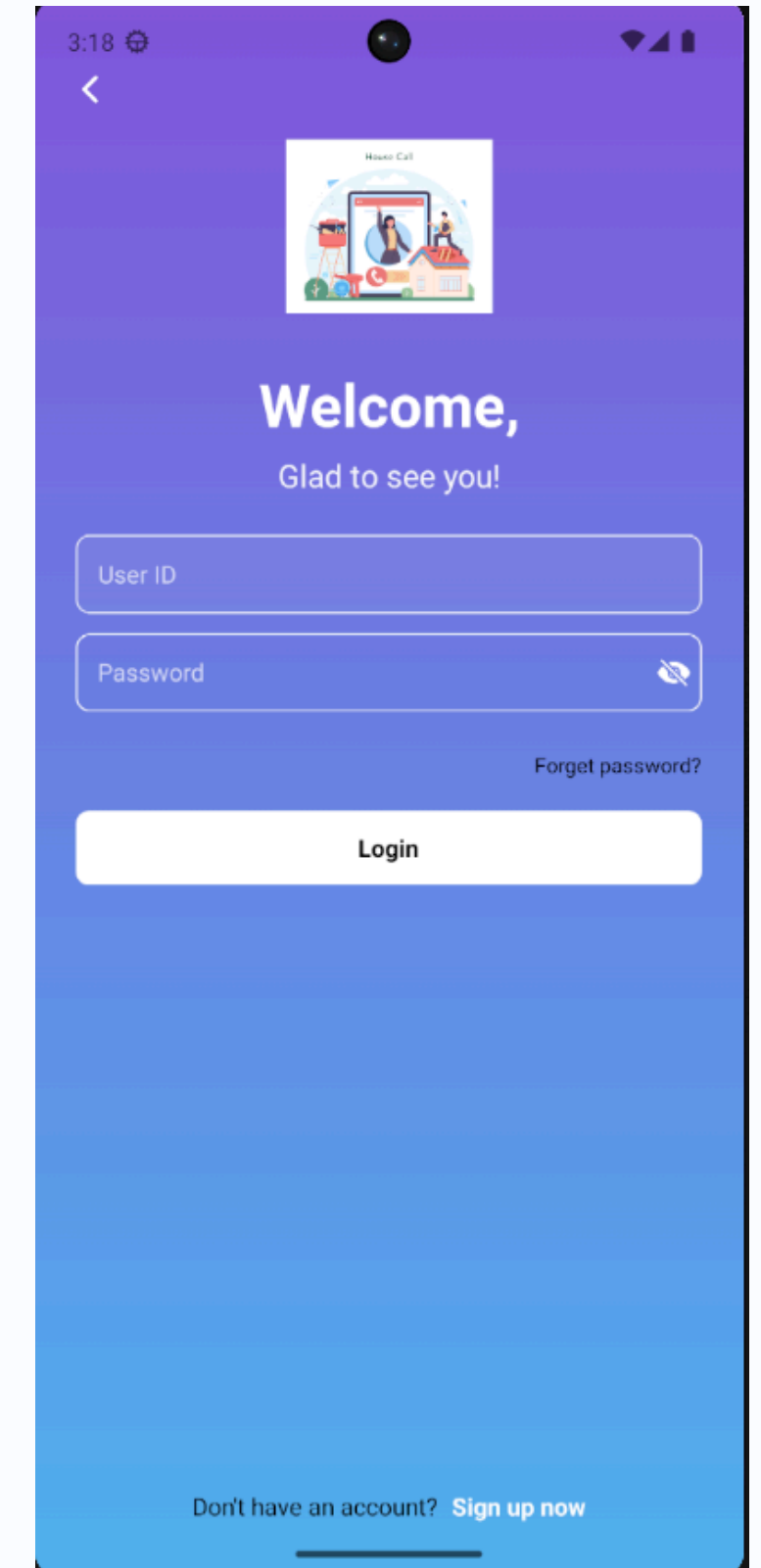
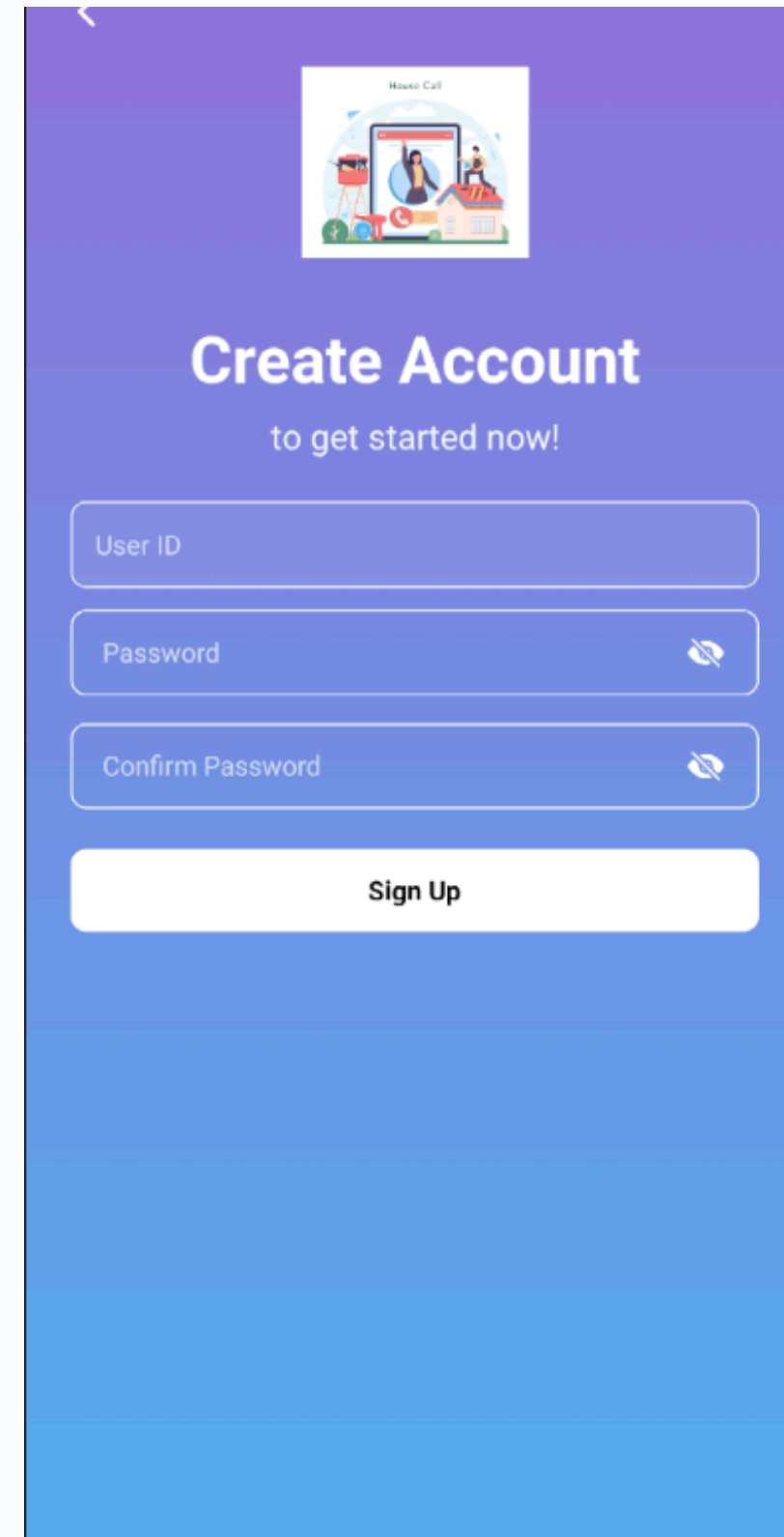
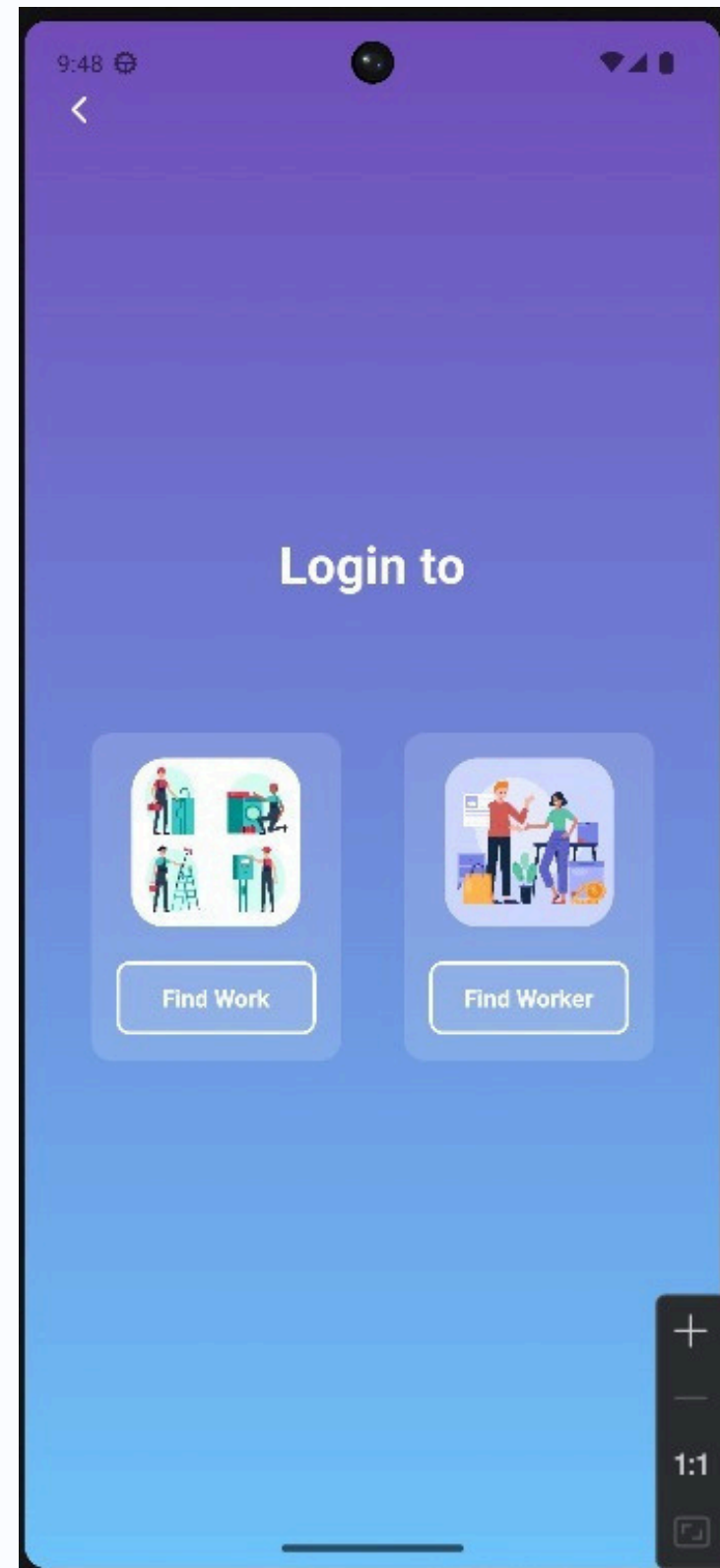
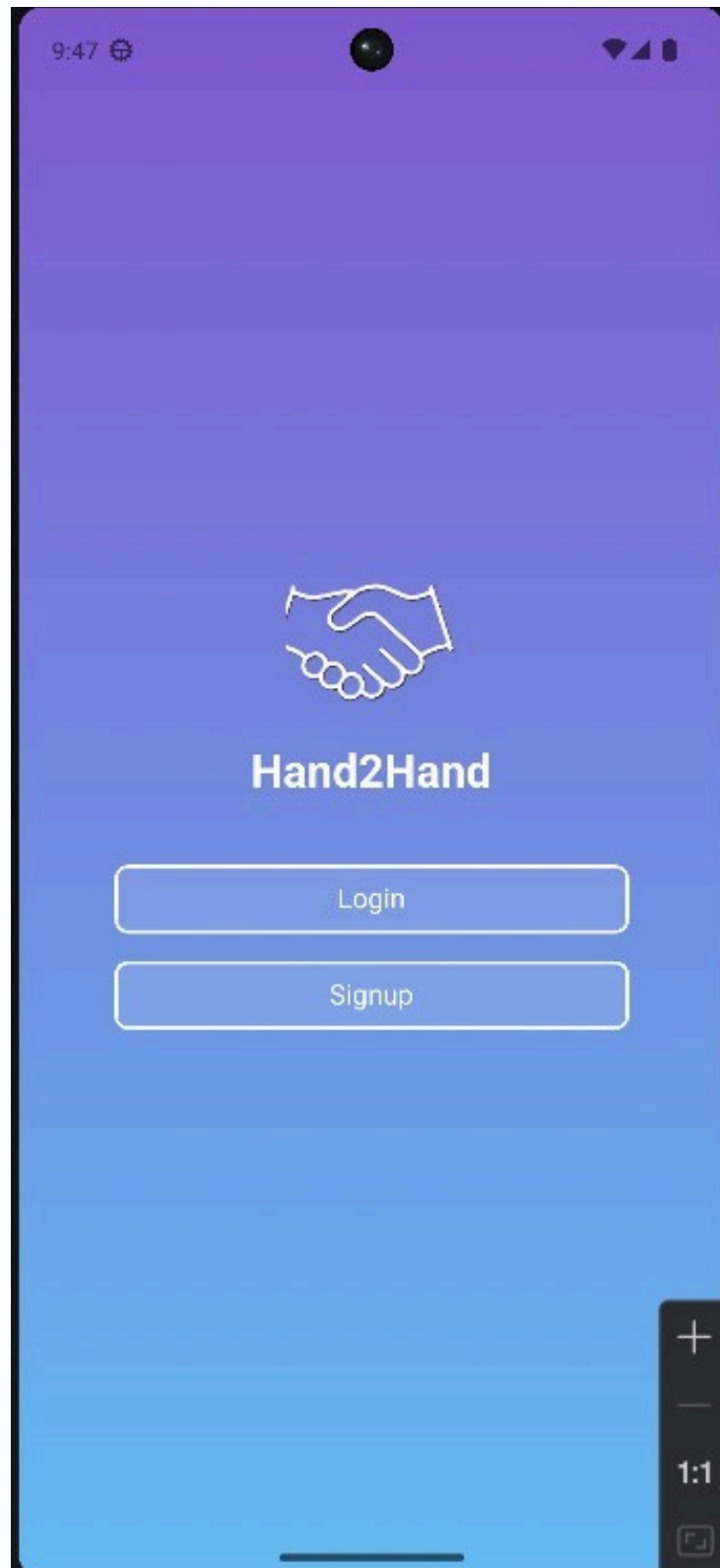
WORKFLOW

Hand2Hand App Workflow

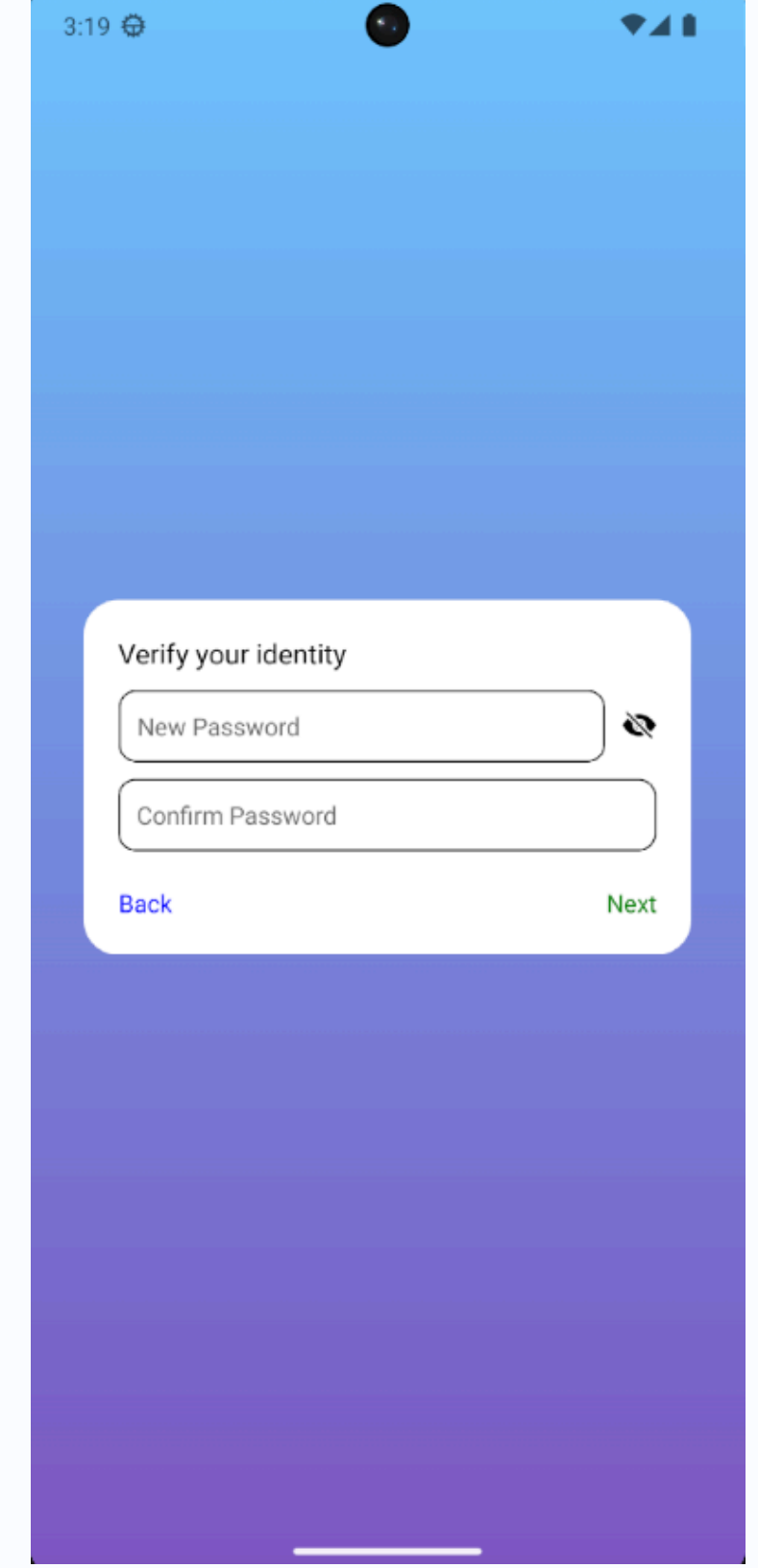
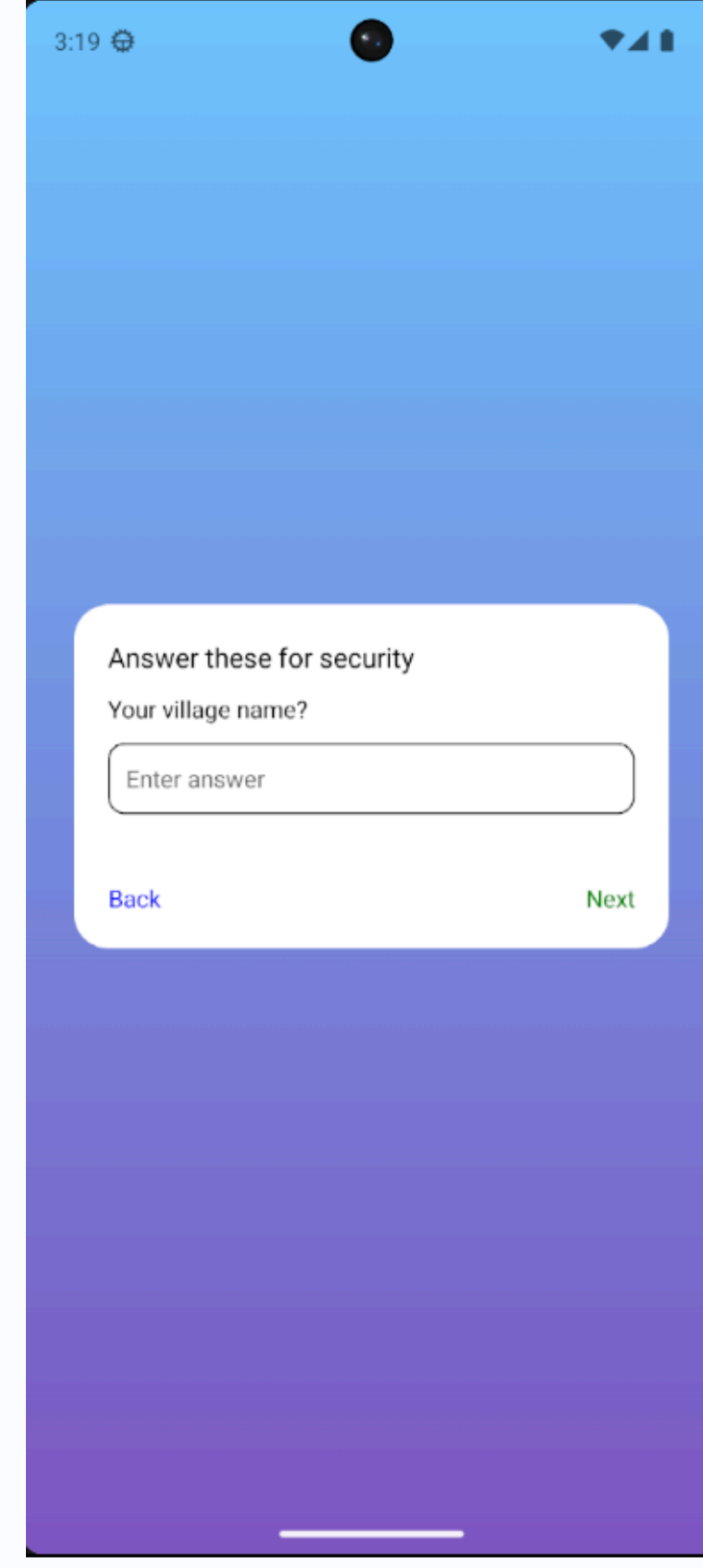
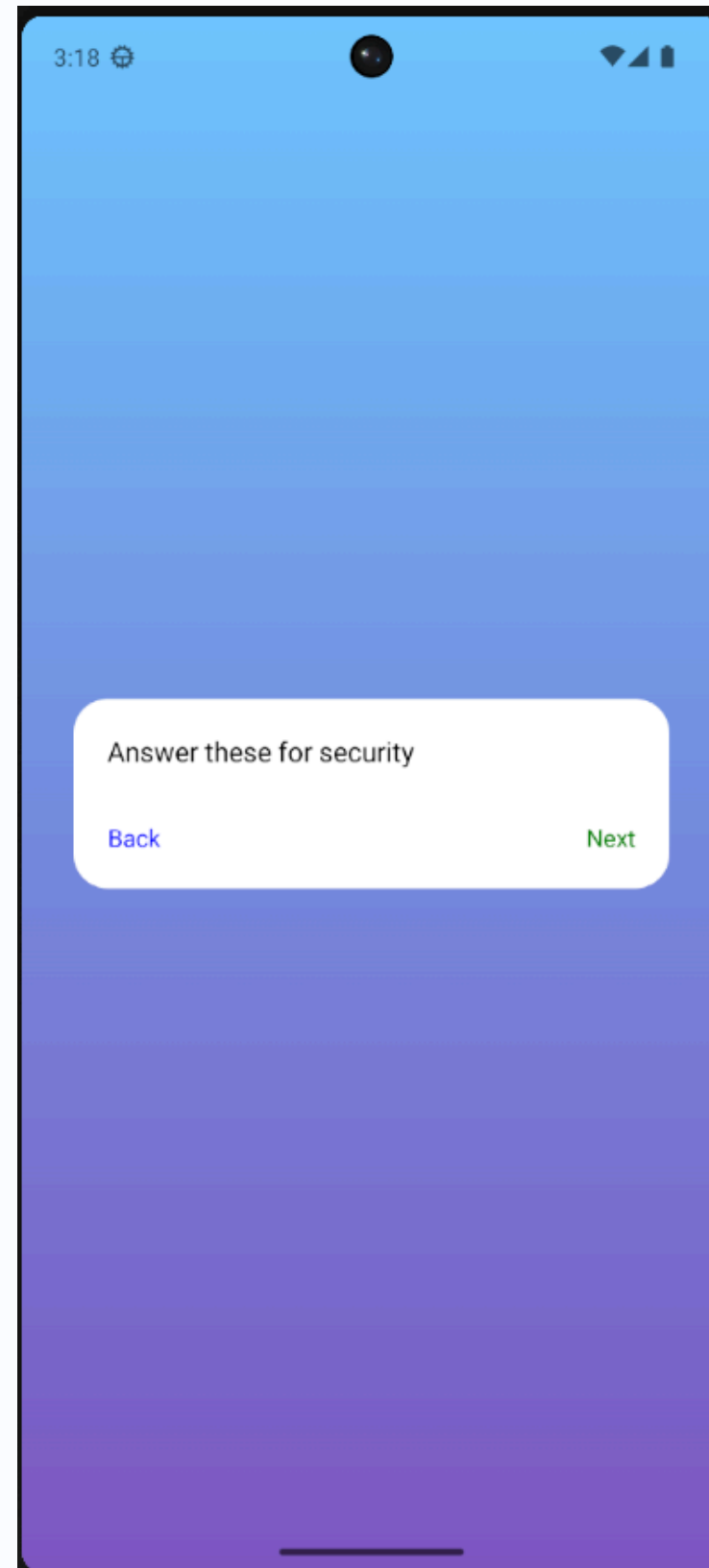
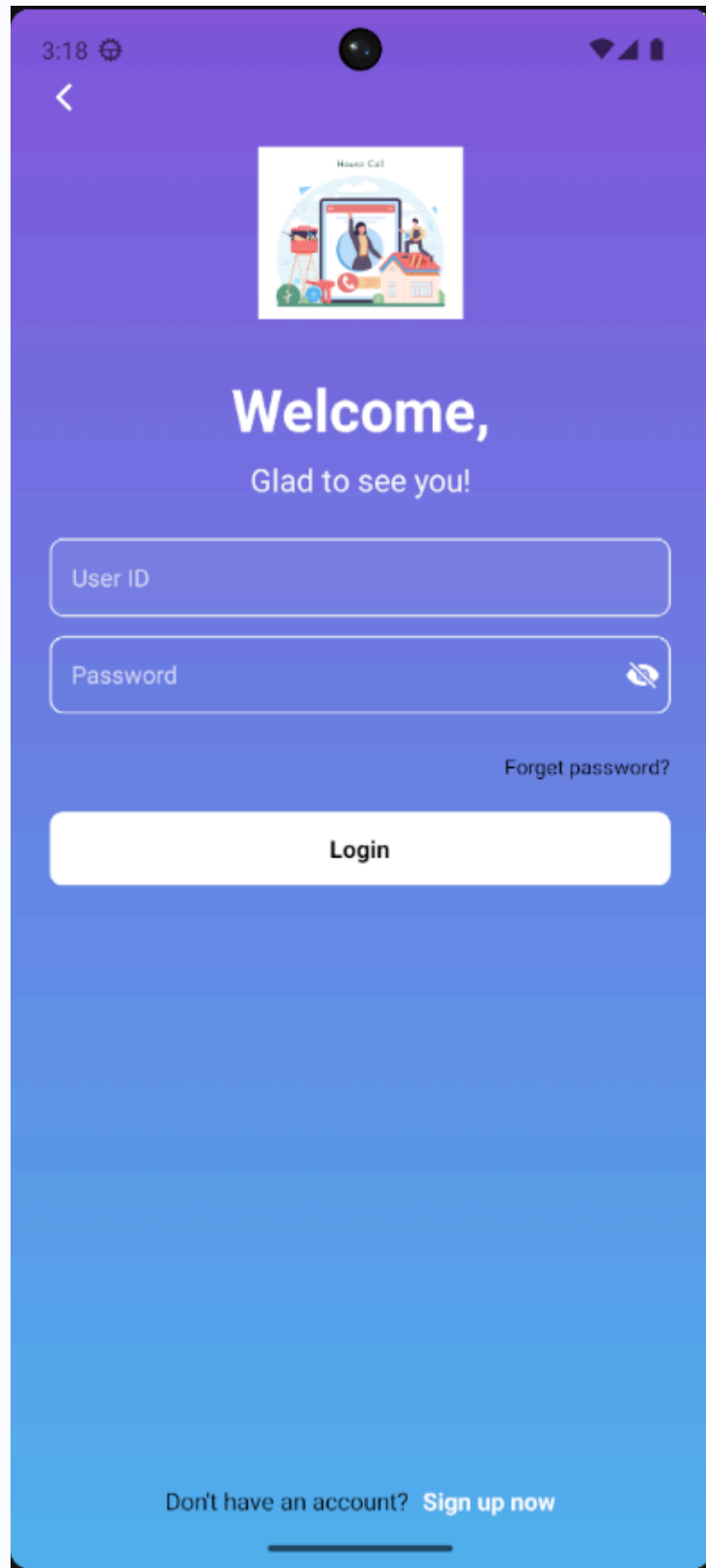


App Workflow Diagram

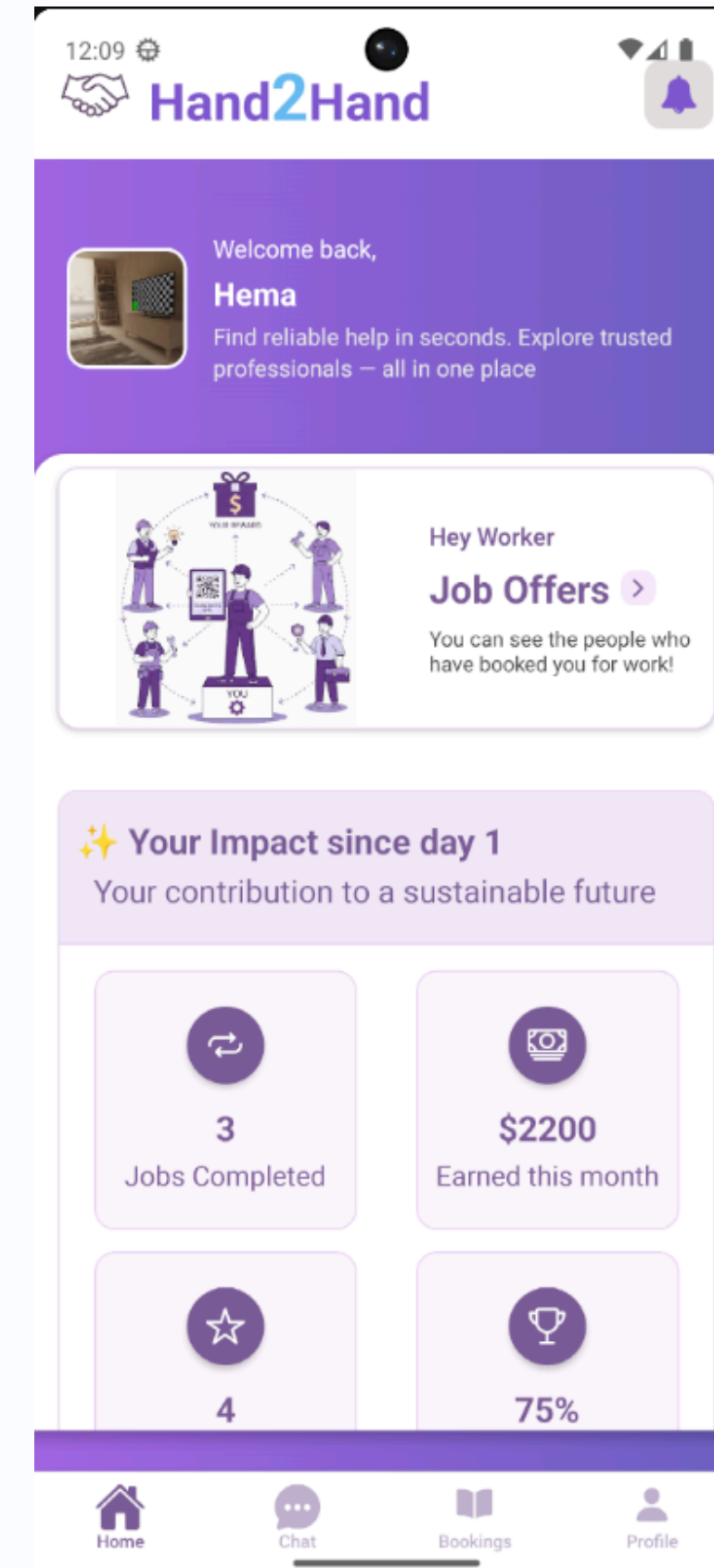
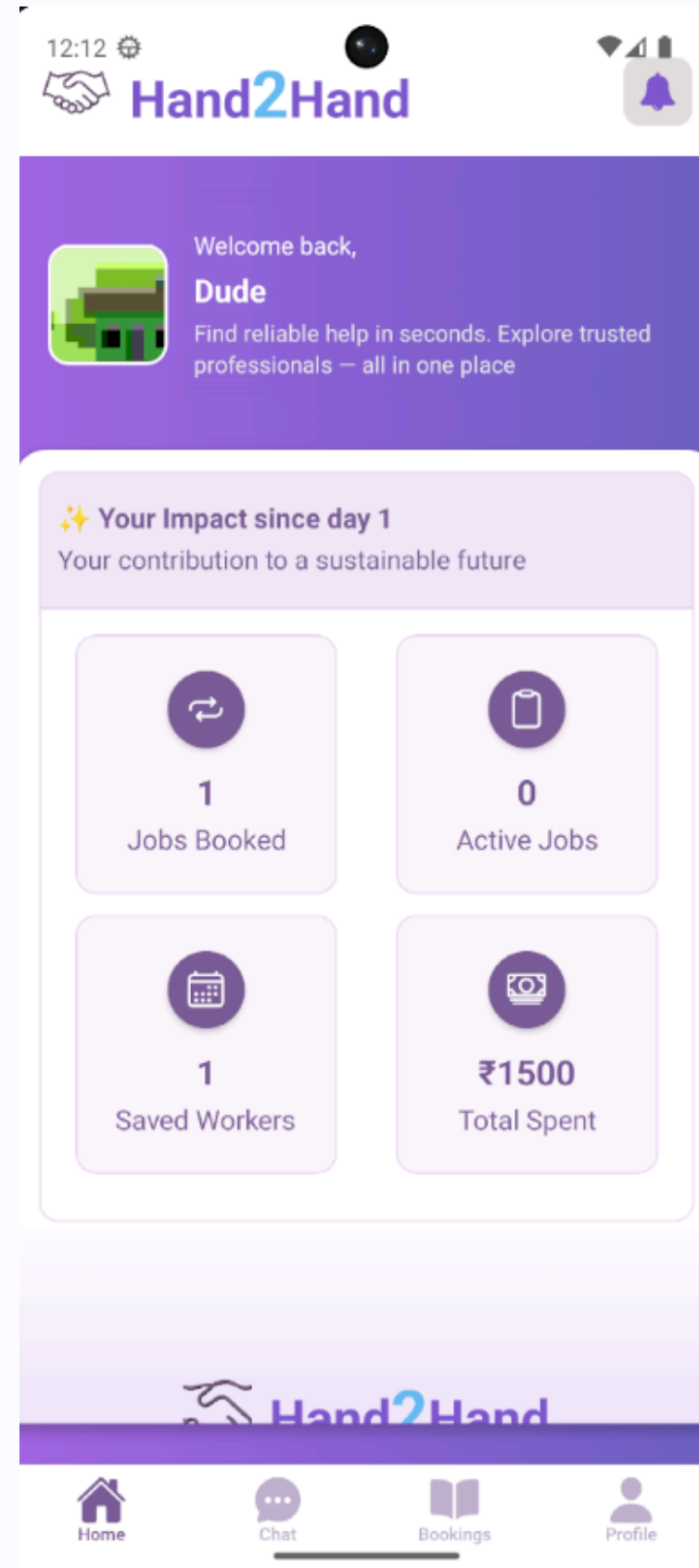
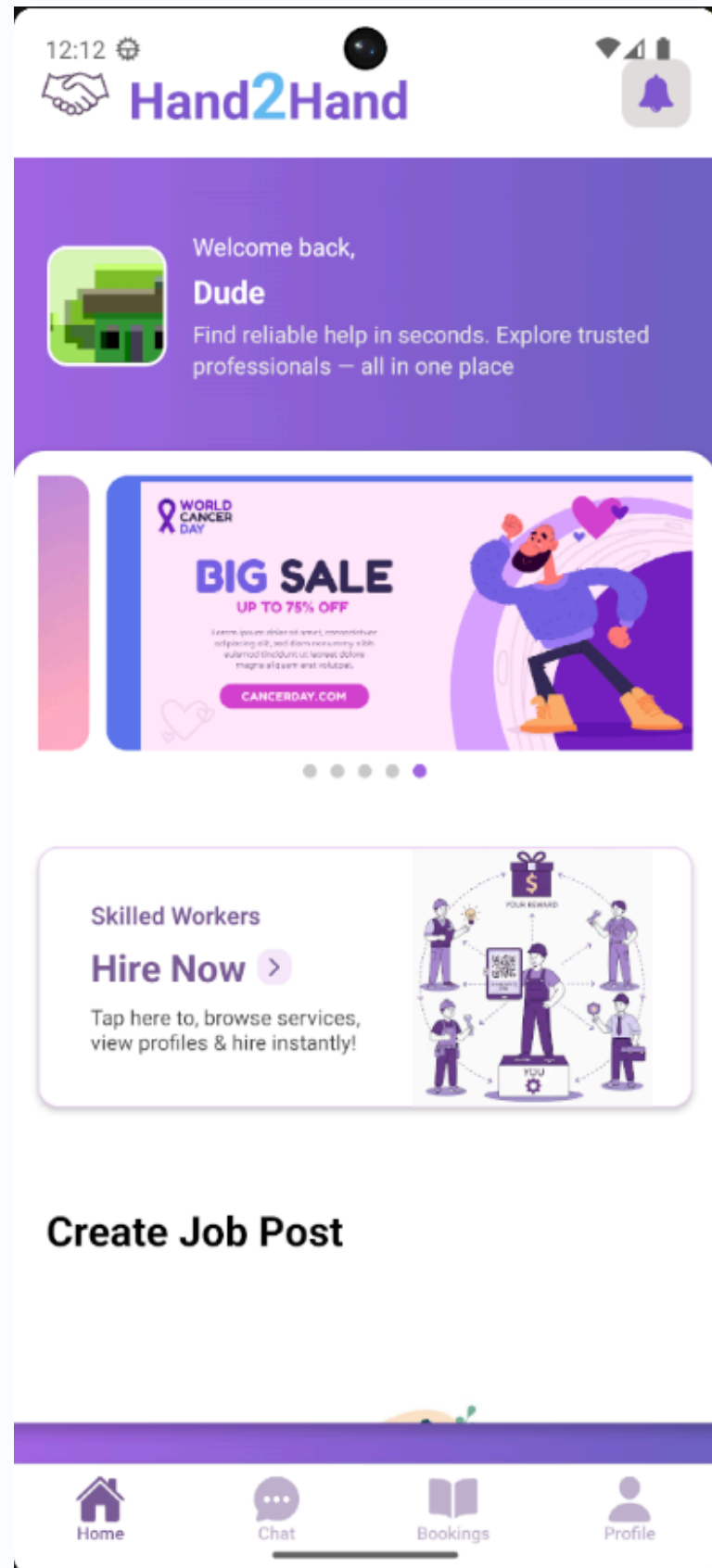
FRONTEND



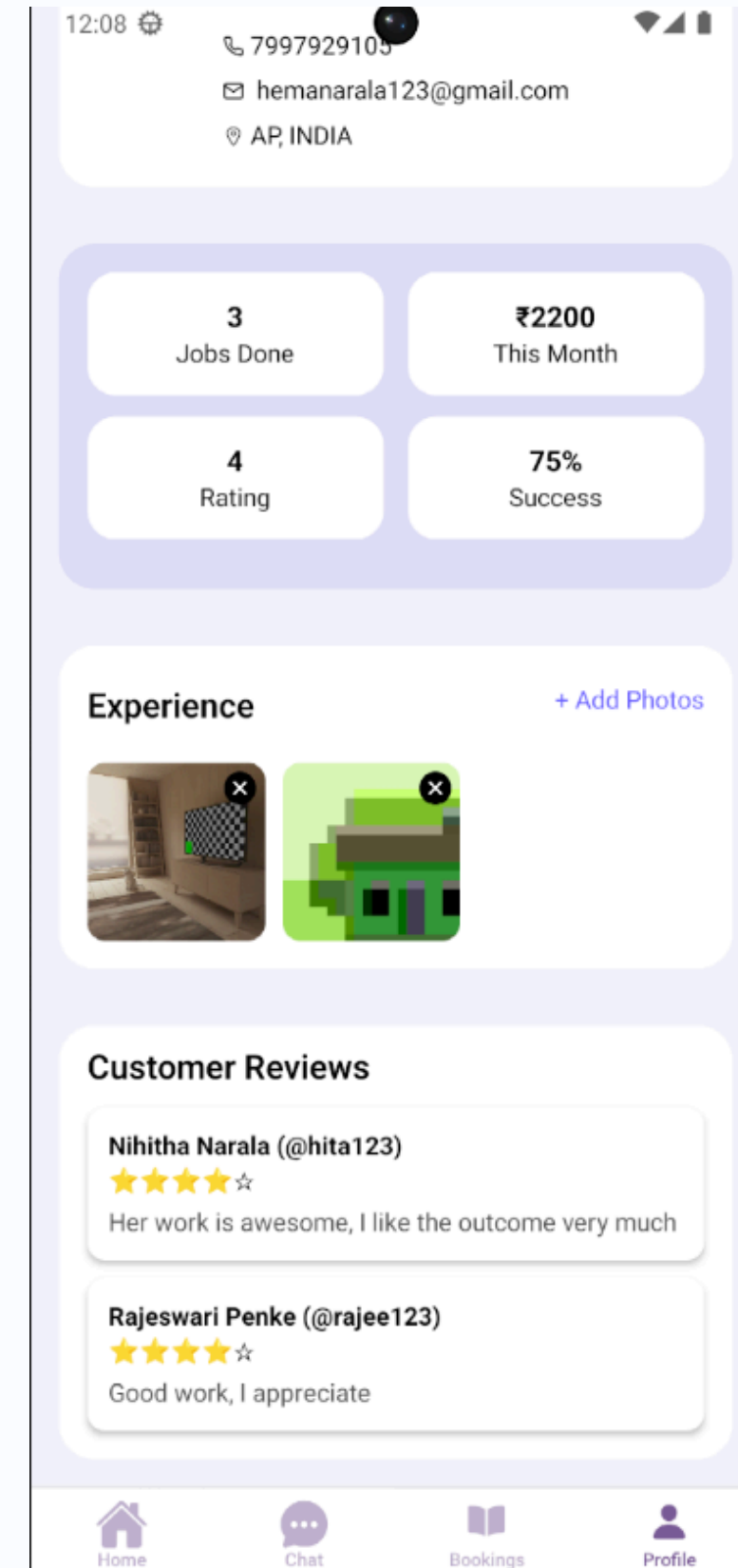
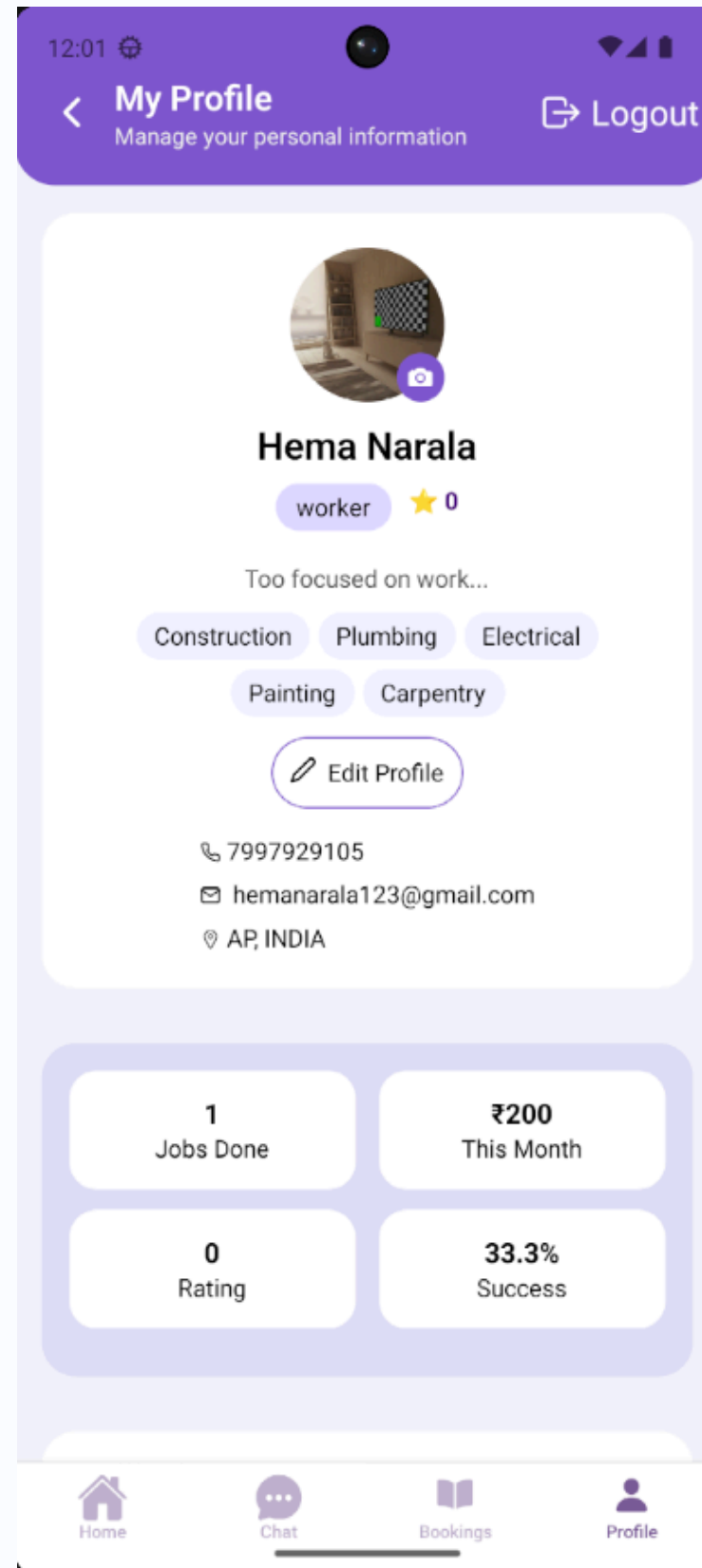
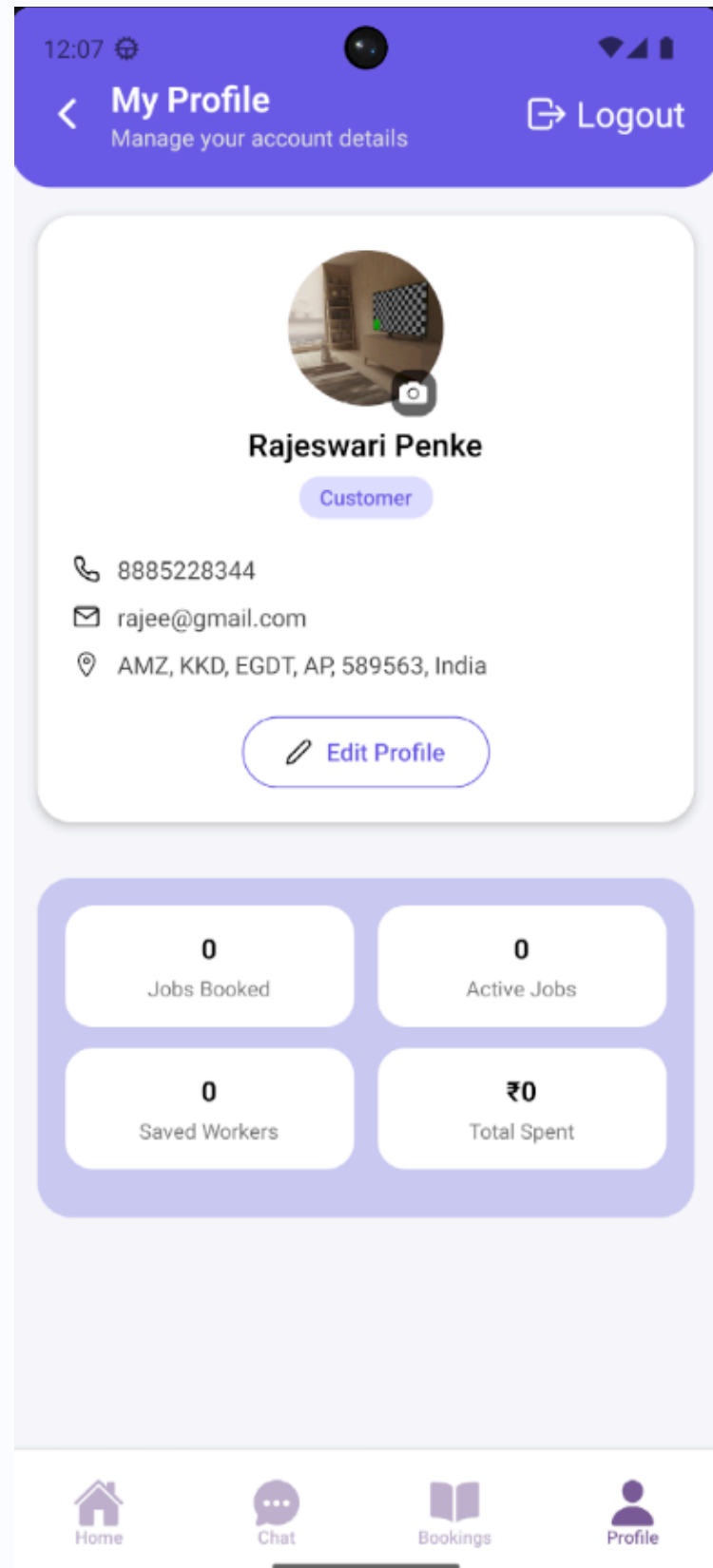
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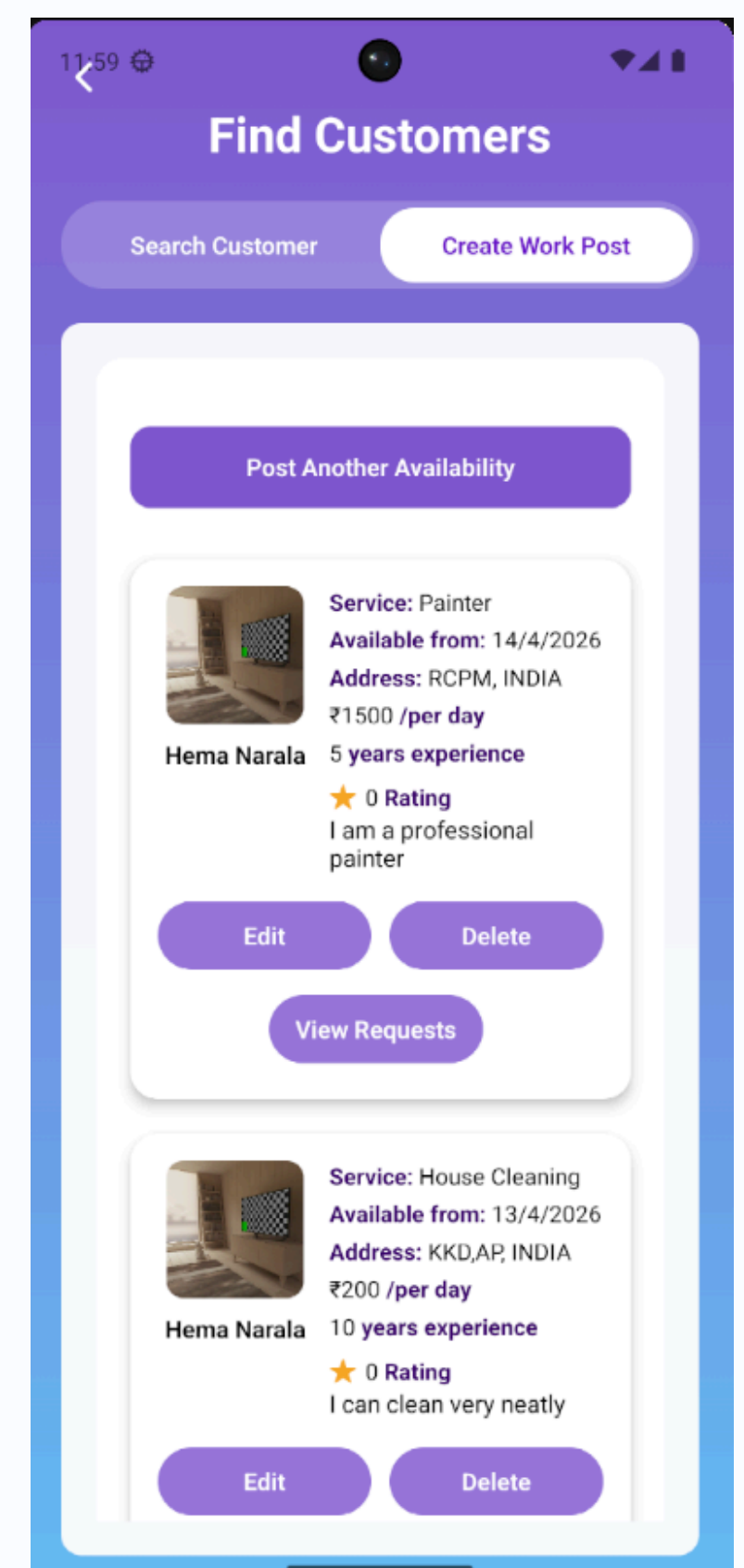
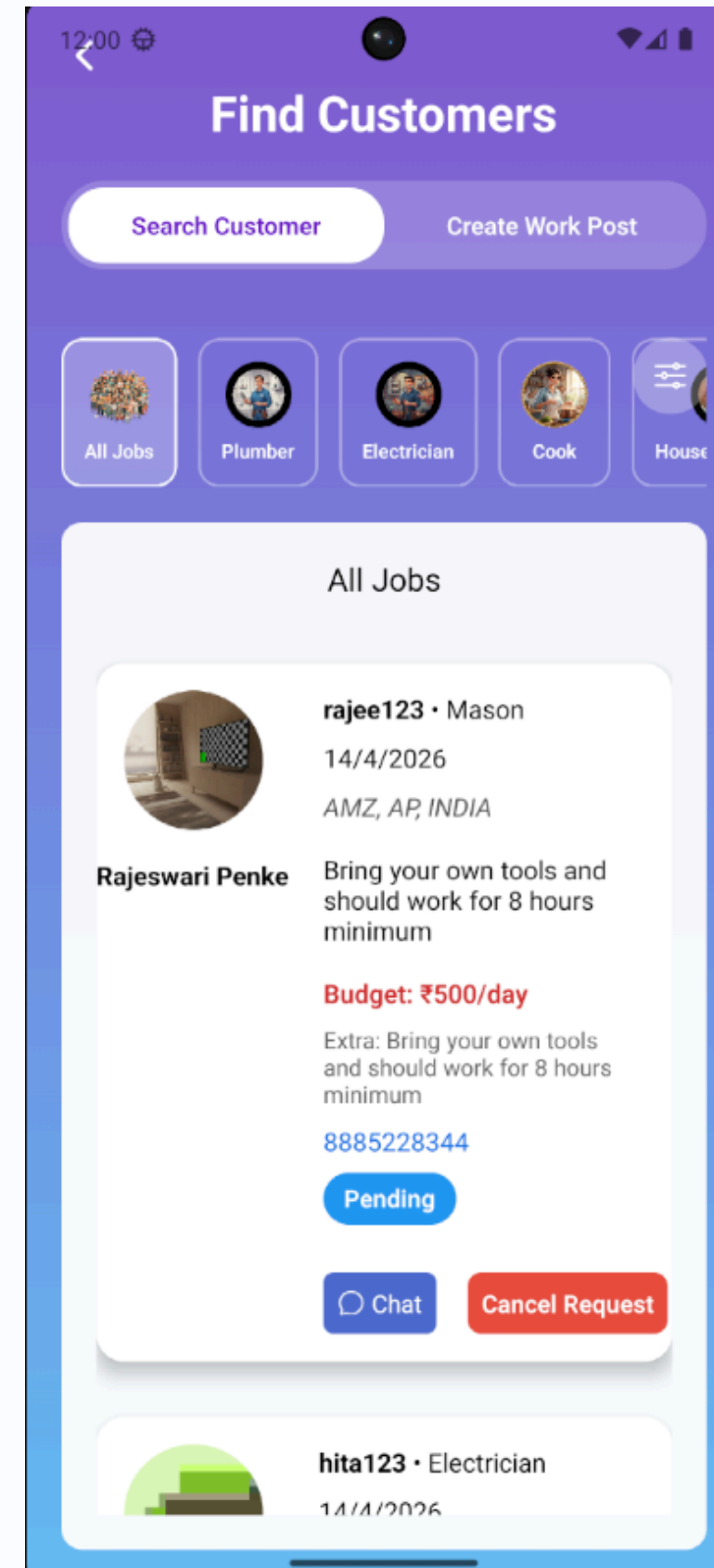
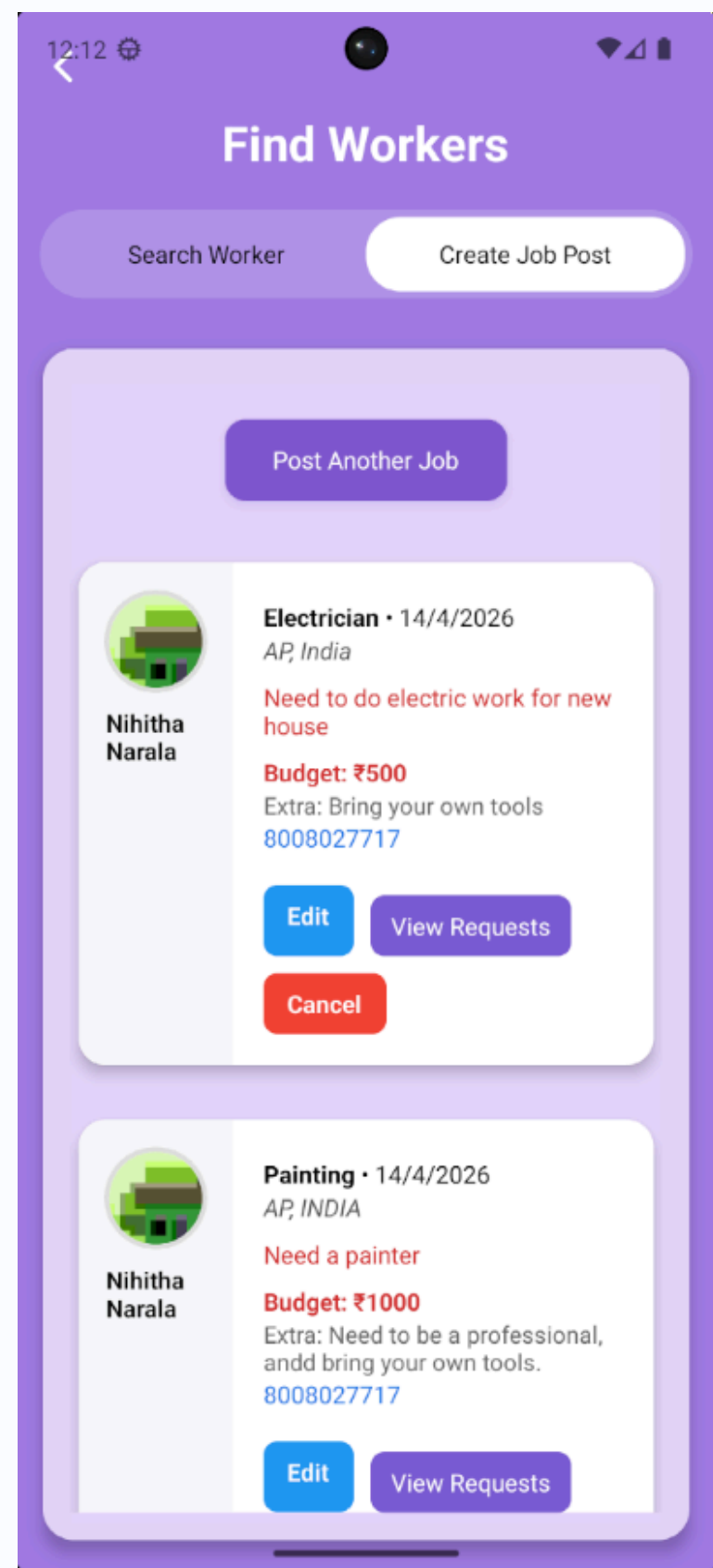
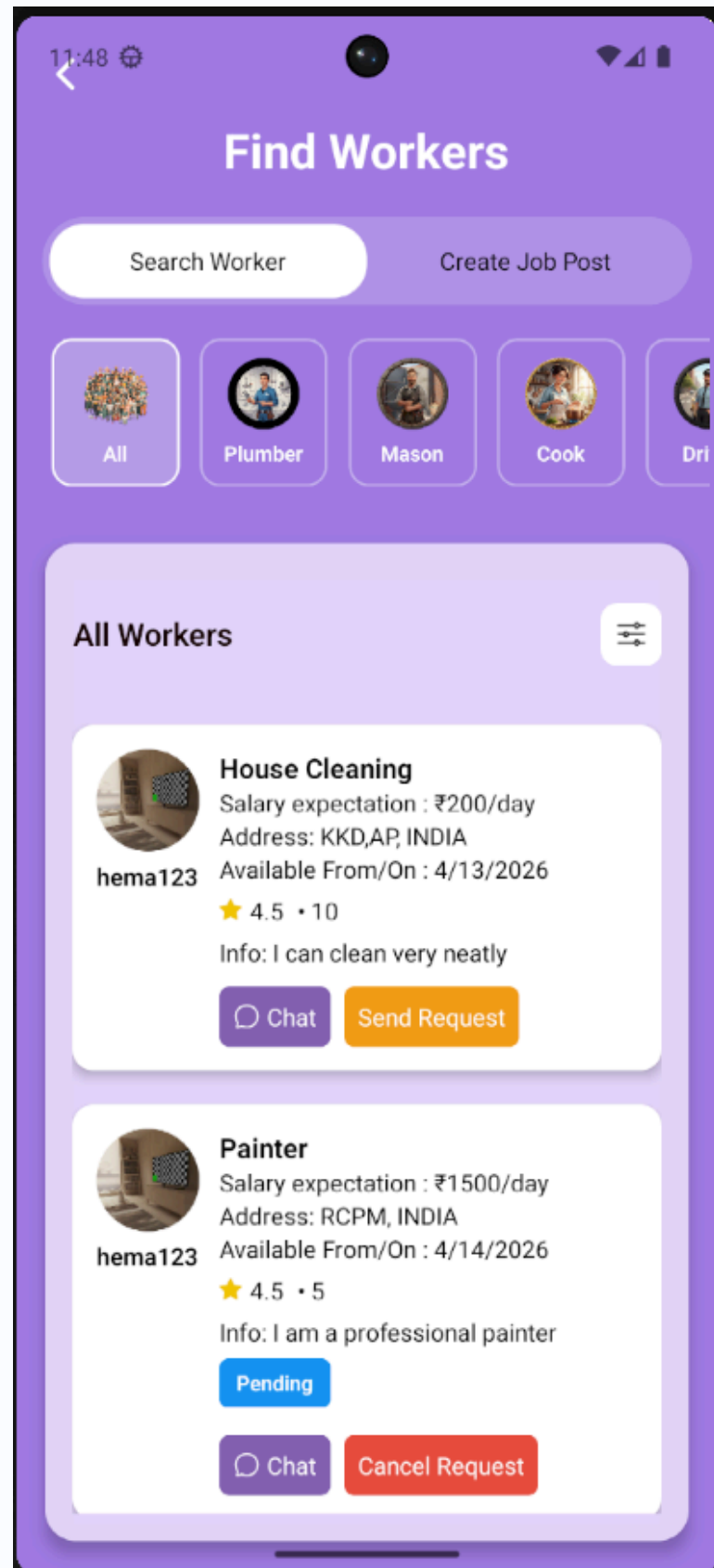
FRONTEND



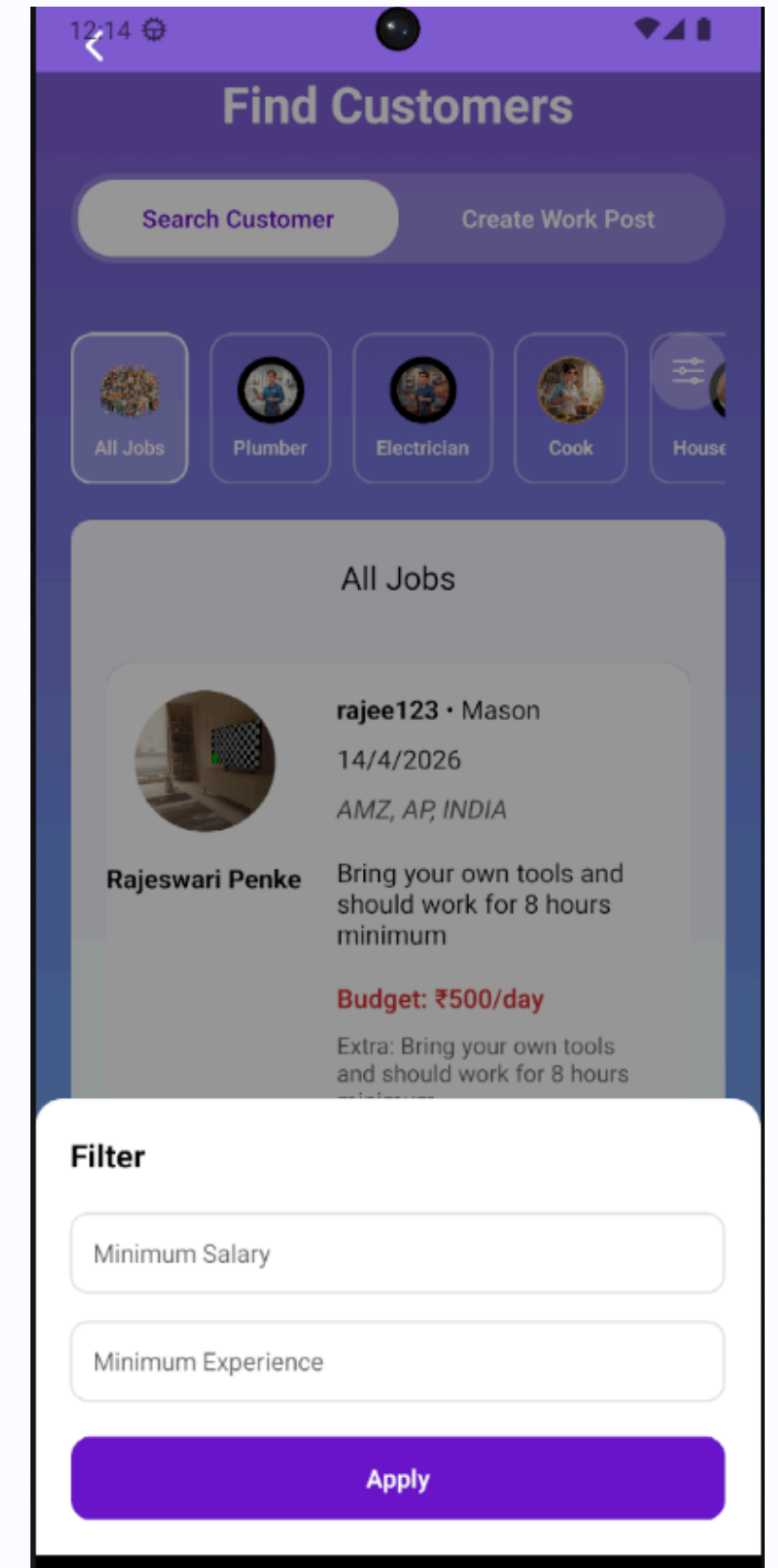
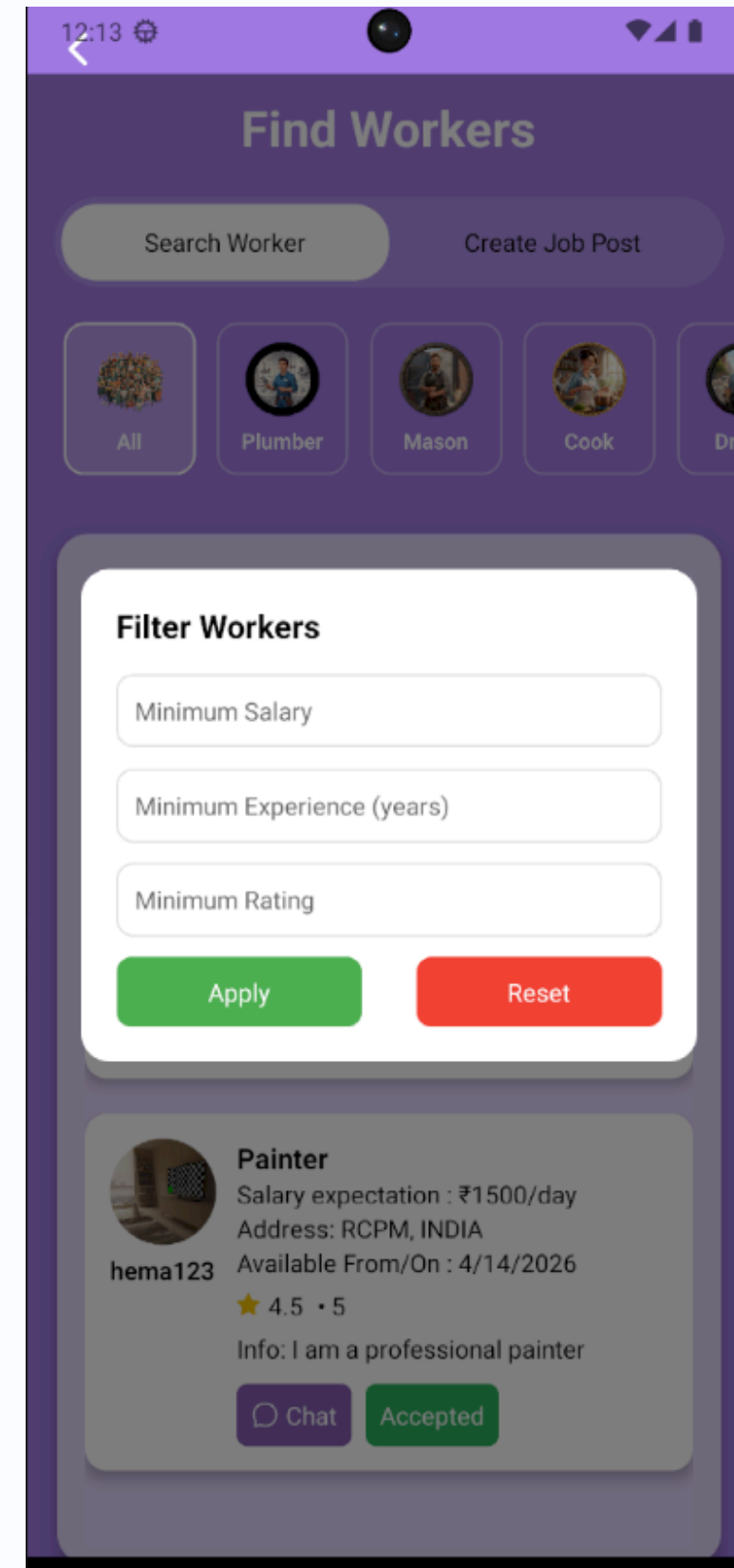
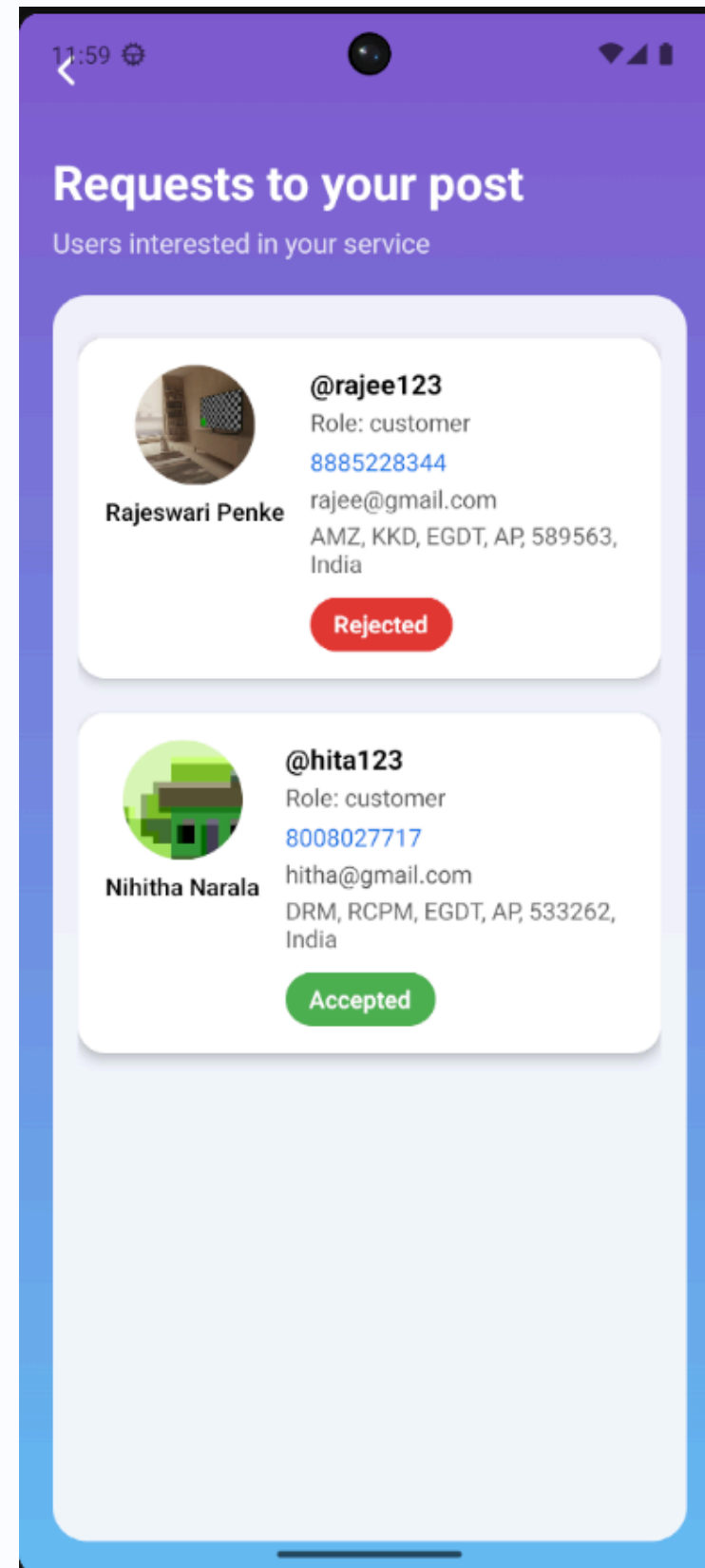
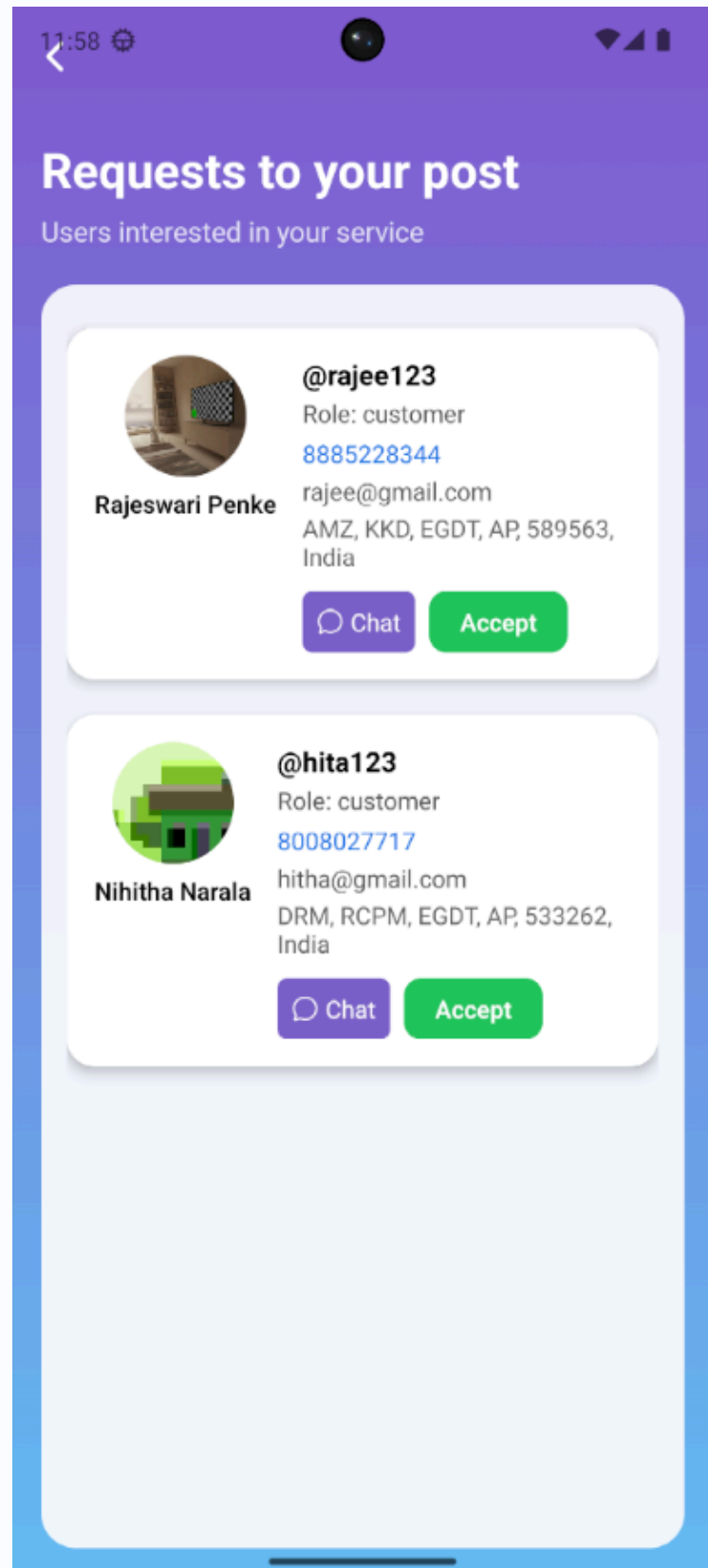
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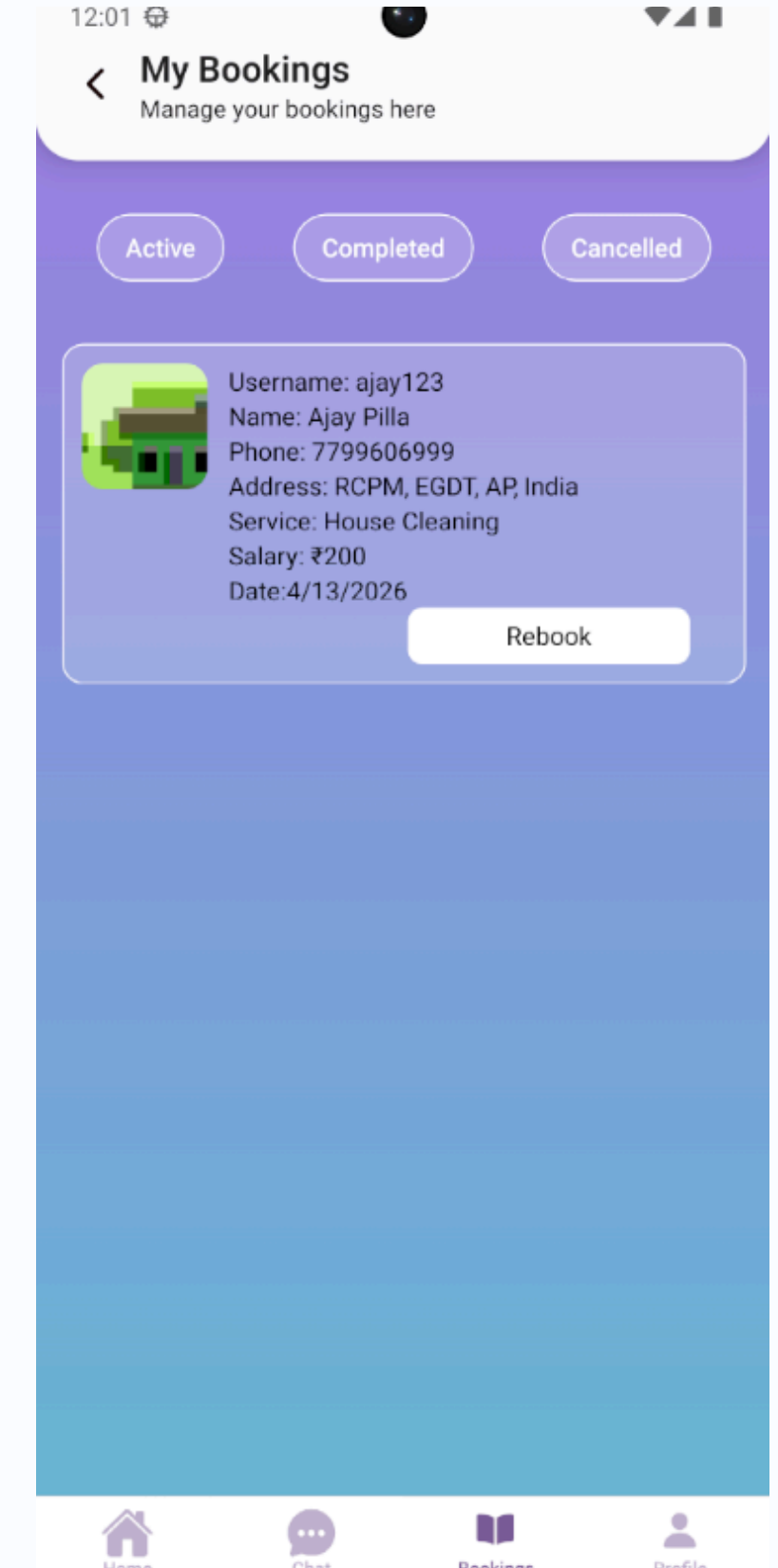
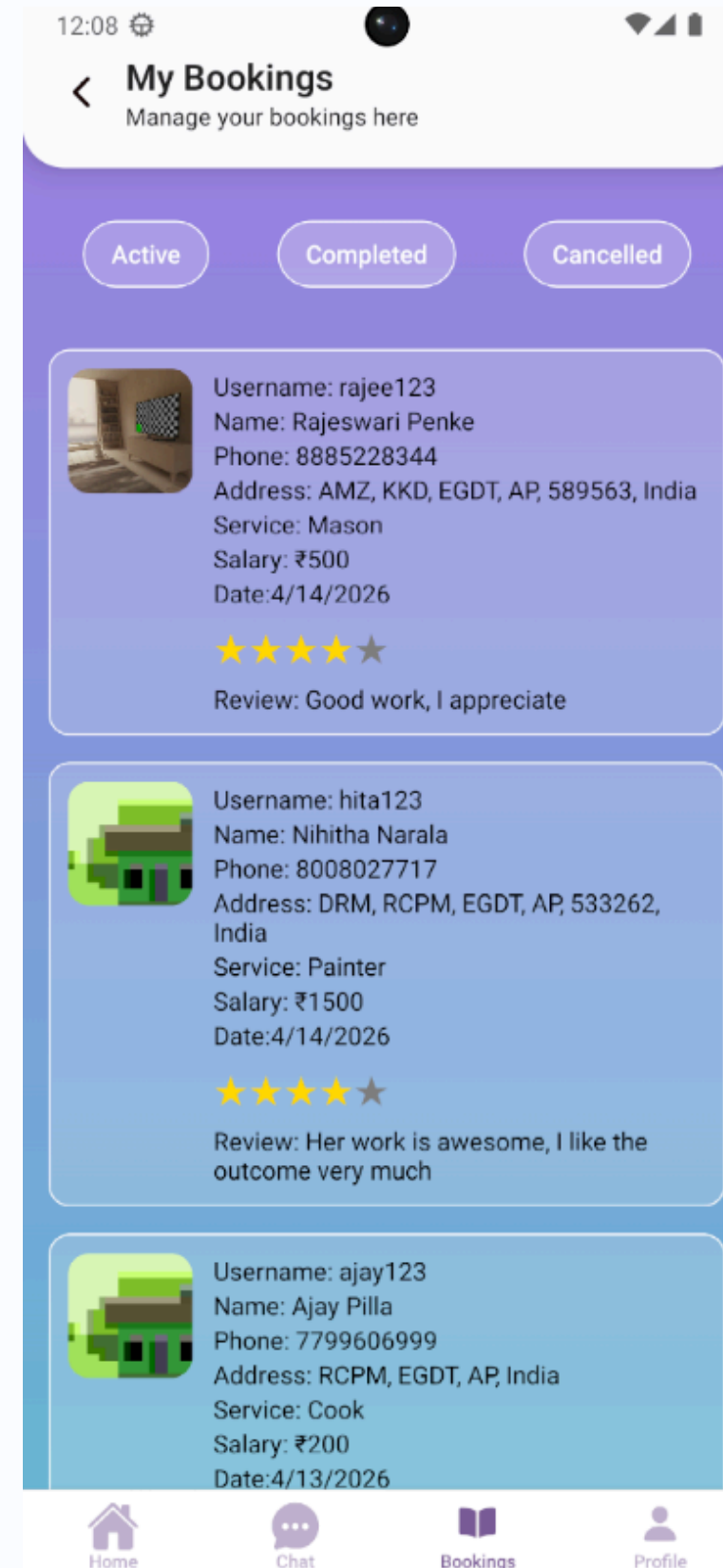
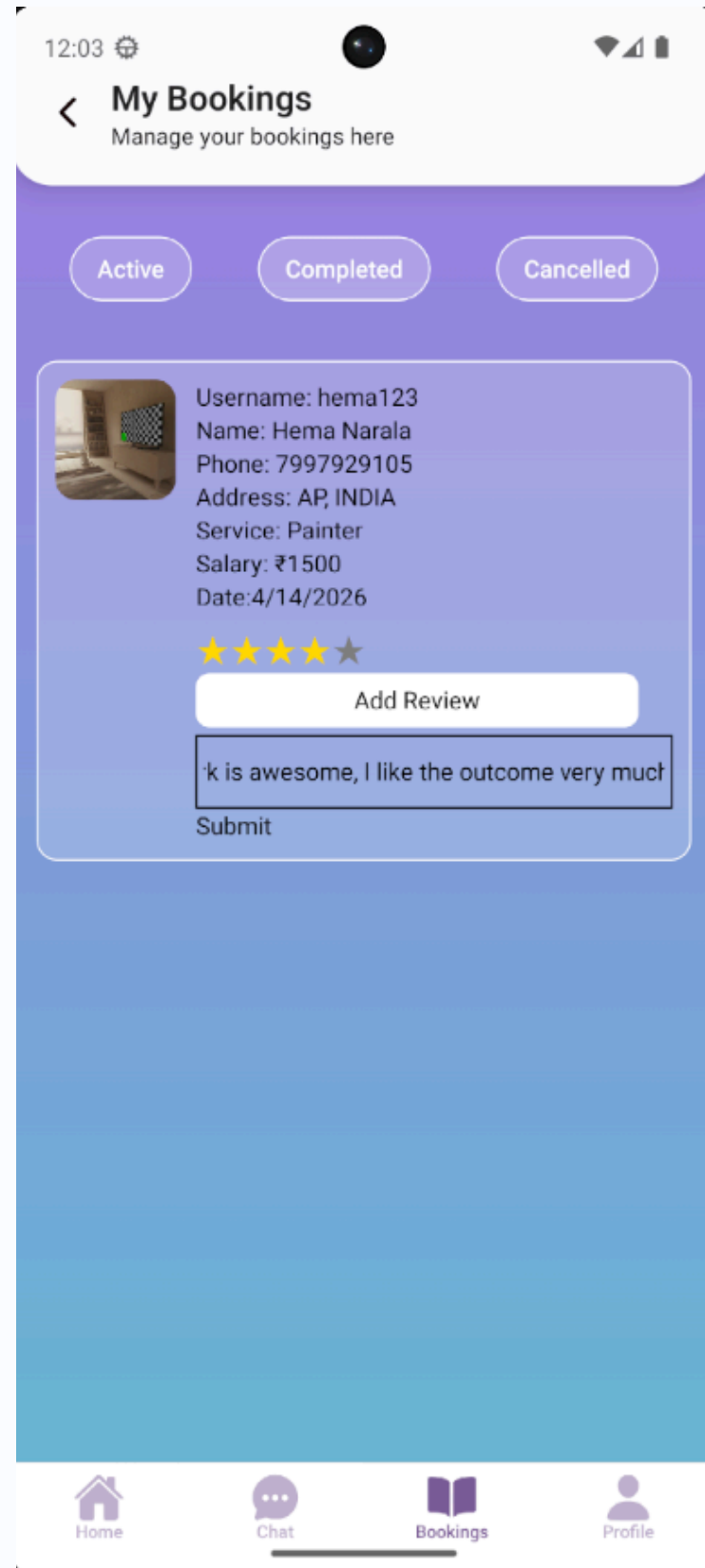
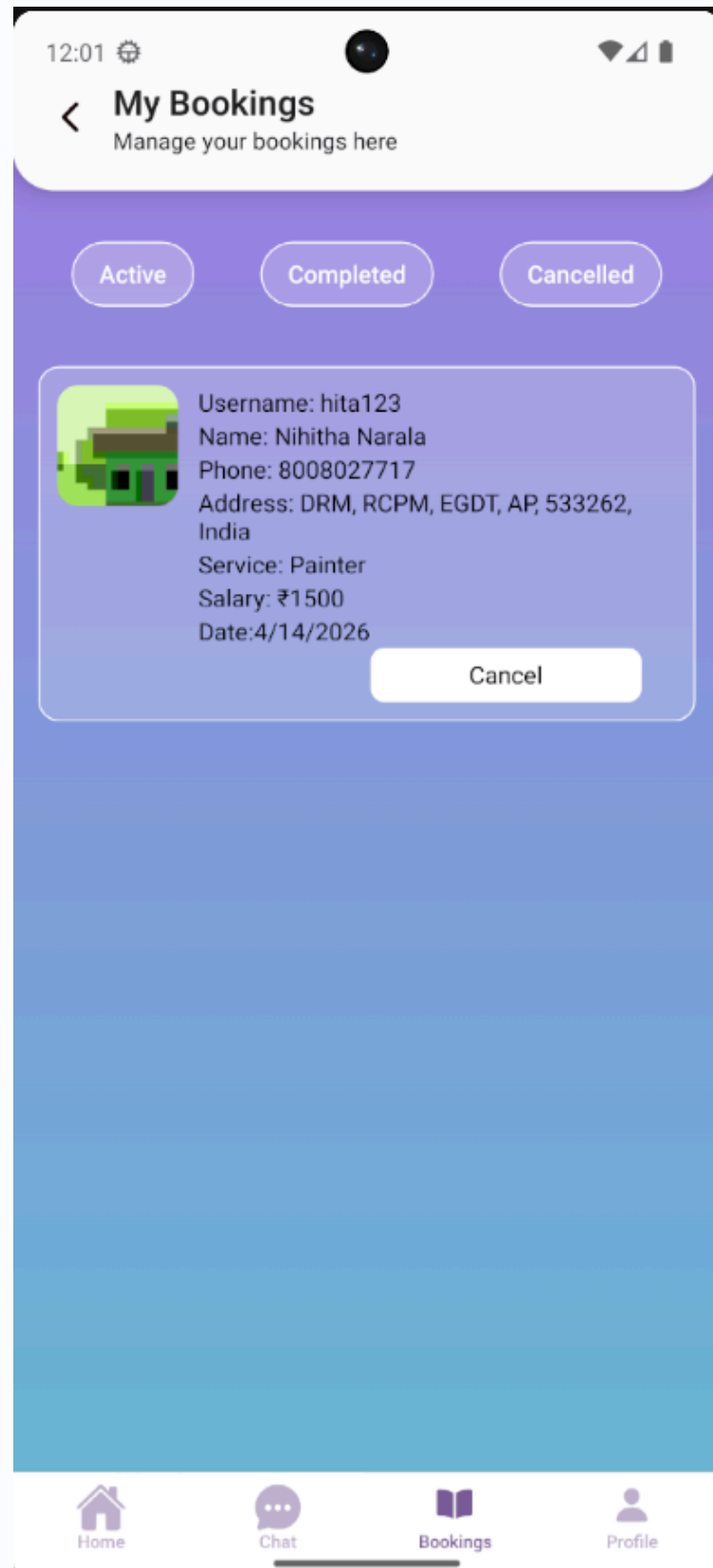
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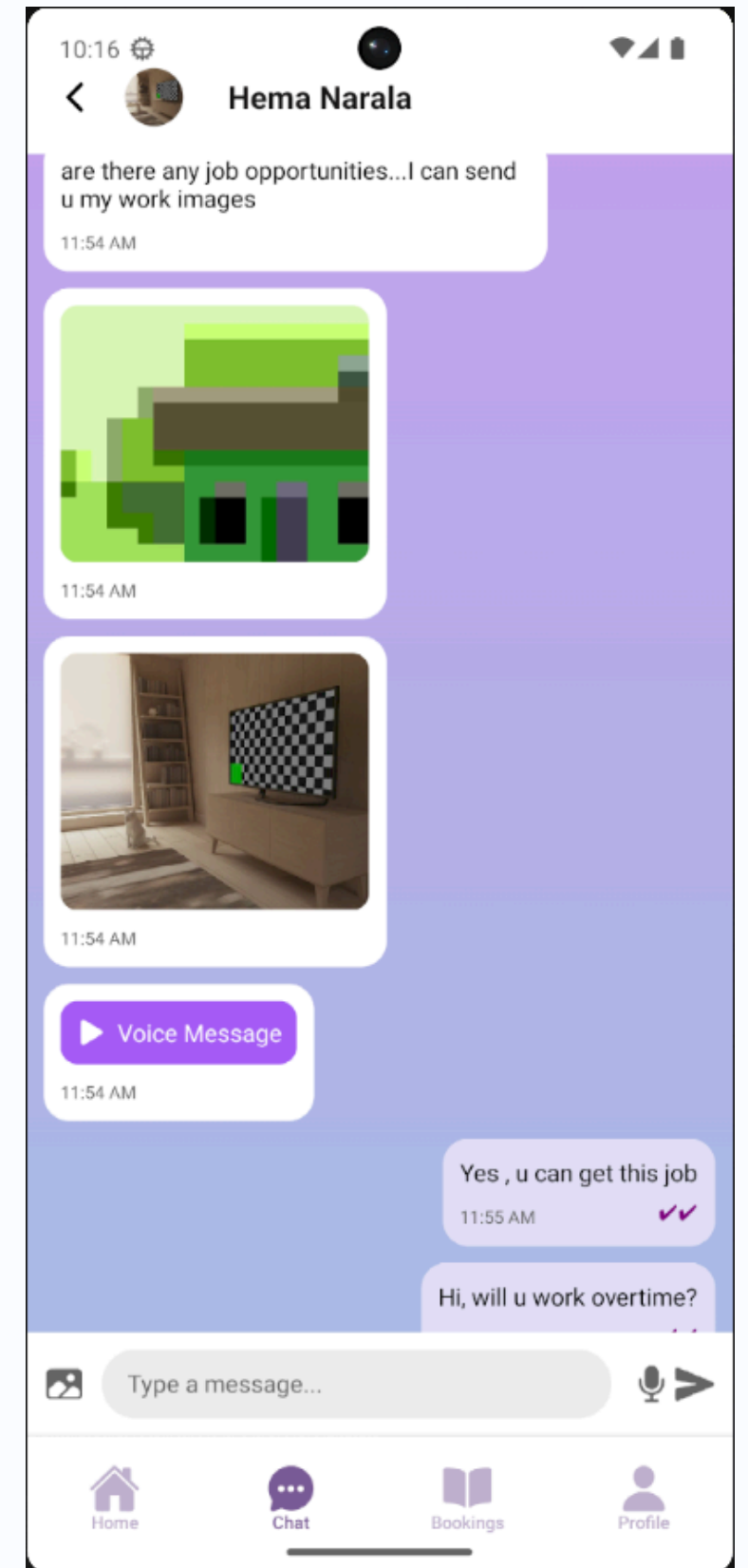
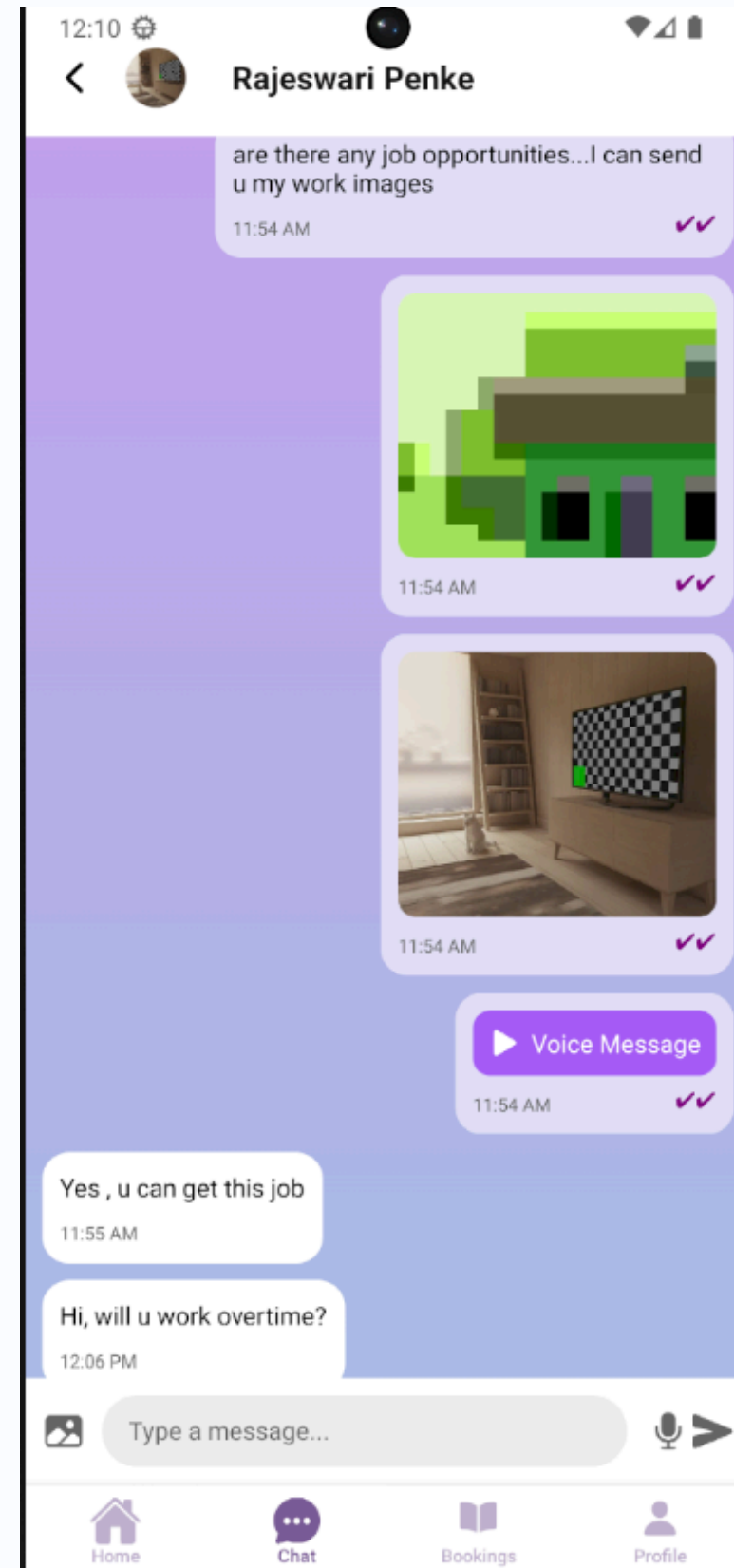
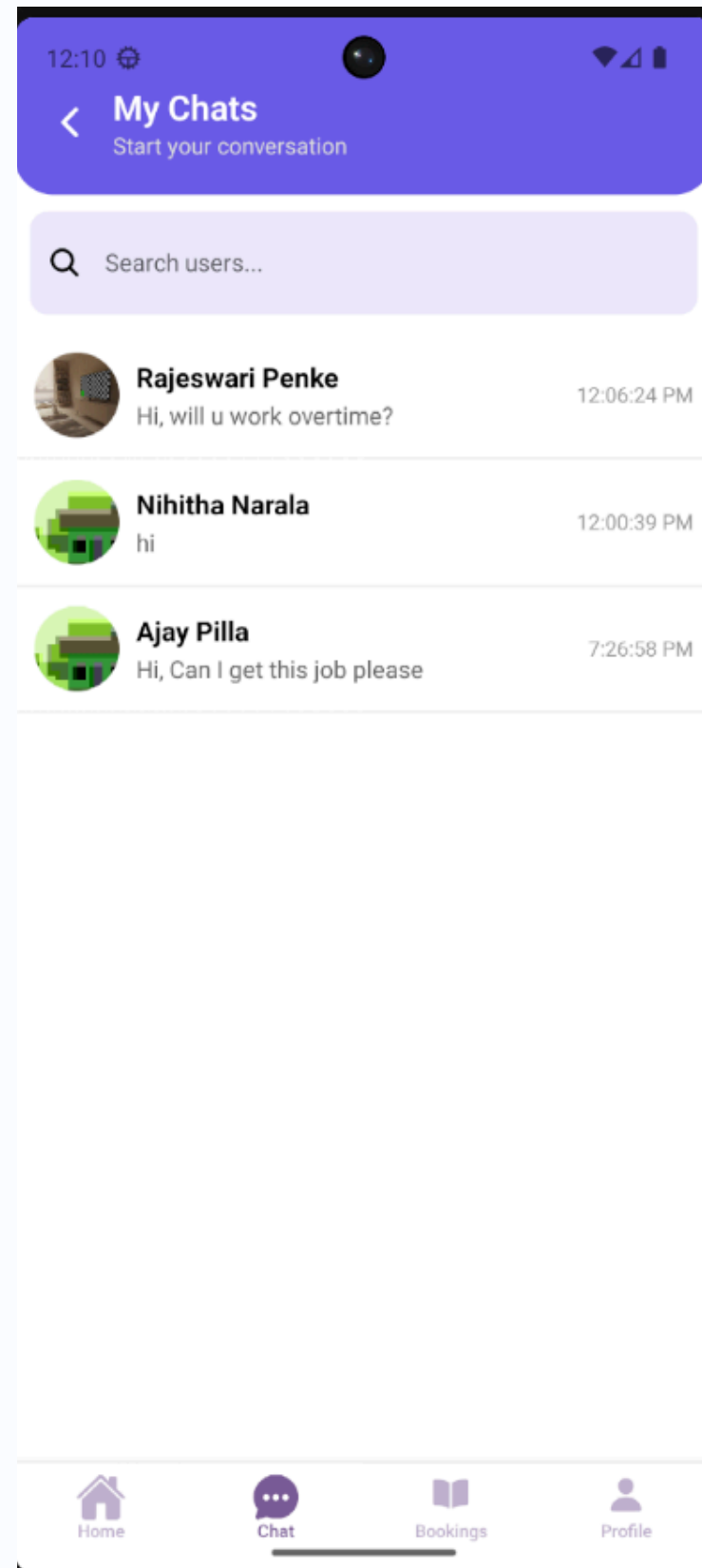
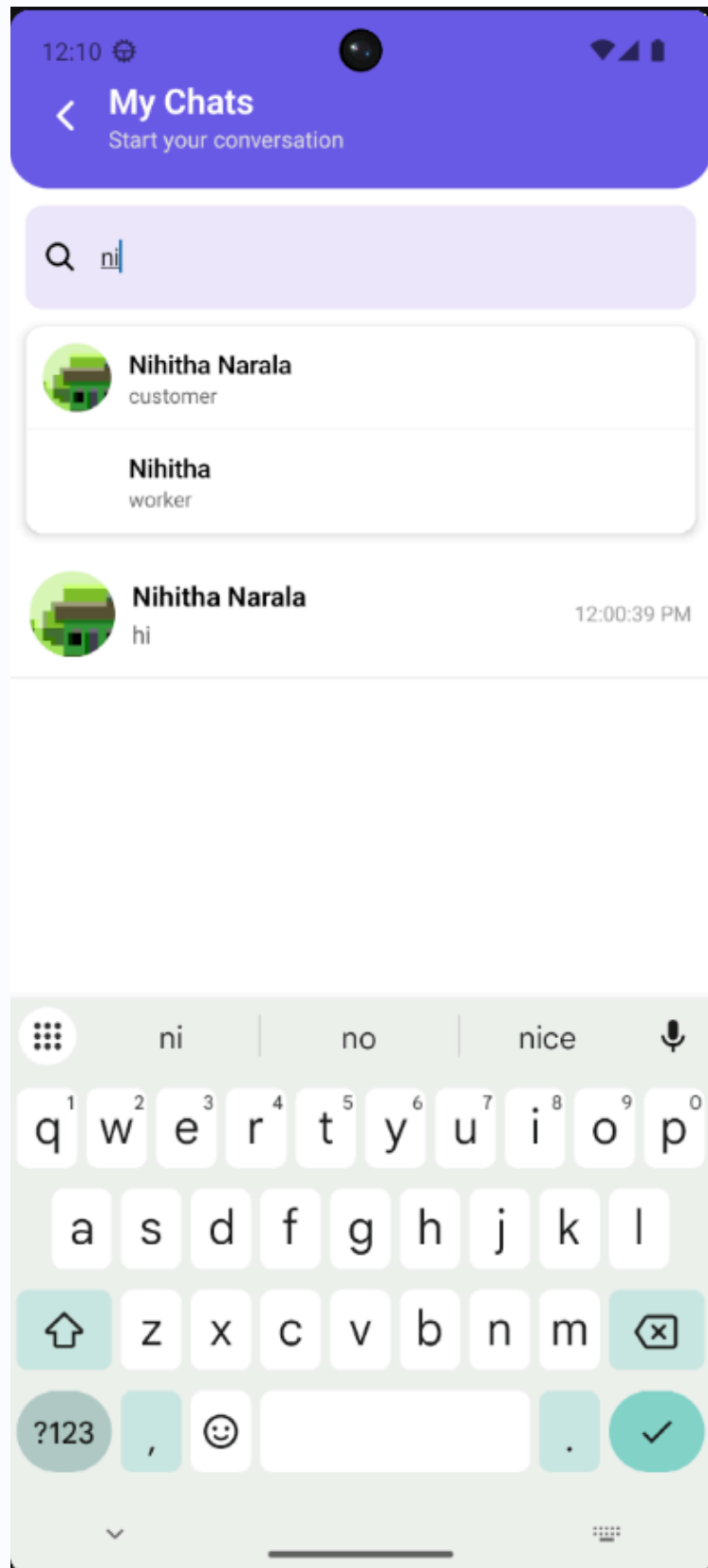
FRONTEND



FRONTEND



FRONTEND



Node Modules

```
EXPLORER
AND2HANDPROJ
hand2hand
  backend
  config
  controllers
  middleware
  models
  node_modules
  routes
  .env
  .gitignore
  package-lock.json
  package.json
  server.js
  frontend
  frontend-clean

JS WorkerProfileScreen.js U
package.json X
JS CustomerProfileScreen.js

hand2hand > backend > package.json > dependencies
1 {
2   "name": "backend",
3   "version": "1.0.0",
4   "description": "",
5   "type": "module",
6   "main": "index.js",
7   "scripts": {
8     "test": "echo \"Error: no test specified\" && exit 1",
9     "start": "node server.js",
10    "dev": "nodemon server.js"
11  },
12  "keywords": [],
13  "author": "",
14  "license": "ISC",
15  "dependencies": {
16    "bcrypt": "^6.0.0",
17    "cloudinary": "^2.9.0",
18    "cors": "^2.8.6",
19    "dotenv": "^17.2.3",
20    "express": "^5.2.1",
21    "jsonwebtoken": "^9.0.3",
22    "mongoose": "^9.1.5",
23    "multer": "^2.0.2",
24    "multer-storage-cloudinary": "^4.0.0",
25    "node-fetch": "^3.3.2",
26    "socket.io": "^4.8.3"
27  },
28  "devDependencies": {
29    "nodemon": "^3.1.11"
30  }
31 }
32
```

```
EXPLORER
AND2HANDPROJ
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json
el.config.js
ex.js
package-lock.json
package.json
end-clean

JS WorkerProfileScreen.js U
package.json M X
JS CustomerProfileScreen.js

hand2hand > frontend > package.json > devDependencies
10 },
11 "dependencies": {
12   "@expo/vector-icons": "^15.0.3",
13   "@react-native-async-storage/async-storage": "2.2.0",
14   "@react-native-community/datetimepicker": "8.4.4",
15   "@react-native-picker/picker": "2.11.1",
16   "@react-navigation/bottom-tabs": "^7.14.0",
17   "@react-navigation/native": "^7.1.28",
18   "@react-navigation/native-stack": "^7.13.0",
19   "expo": "~54.0.33",
20   "expo-av": "~16.0.8",
21   "expo-file-system": "~19.0.21",
22   "expo-font": "~14.0.11",
23   "expo-image-picker": "~17.0.10",
24   "expo-linear-gradient": "~15.0.8",
25   "expo-location": "~19.0.8",
26   "expo-status-bar": "~3.0.9",
27   "react": "19.1.0",
28   "react-native": "0.81.5",
29   "react-native-gesture-handler": "~2.28.0",
30   "react-native-reanimated": "~4.1.1",
31   "react-native-safe-area-context": "~5.6.0",
32   "react-native-screens": "~4.16.0",
33   "socket.io": "^4.8.3",
34   "socket.io-client": "^4.8.3"
35 },
36 "private": true,
37 "devDependencies": {
38   "babel-preset-expo": "^54.0.10"
39 }
40 }
```

Database Using MongoDB

The screenshot displays the MongoDB Atlas web interface. On the left sidebar, under 'CLUSTERS (1)', 'Cluster0' is expanded, showing a hierarchy of databases: 'admin', 'hand2hand', and 'local'. The 'hand2hand' database is selected, and its collections are listed: 'bookings', 'chatrooms', 'customerposts', 'customerprofiles', 'messages', 'notifications', 'products', 'requests', 'users', 'workerposts', and 'workerprofiles'. The 'users' collection is highlighted.

The main panel shows the 'Cluster0 > hand2hand > users' view. It has tabs for 'Documents' (29), 'Aggregations', 'Schema', 'Indexes' (2), and 'Validation'. The 'Documents' tab is active, displaying a search bar with the placeholder 'Type a query: { field: 'value' } or [Generate query](#)'. Below the search bar are buttons for 'ADD DATA', 'UPDATE', 'DELETE', and 'EXPORT CODE'.

The document list shows two documents. The first document is expanded, showing its structure:

- securityQuestions: Array (3)
- address: Object
- skills: Array (empty)
- rating: 0
- points: 0
- bookings: Array (empty)
- createdAt: 2026-04-14T05:31:58.687+00:00
- updatedAt: 2026-04-14T05:33:13.703+00:00
- __v: 0

The second document is also expanded, showing its structure:

- _id: ObjectId('69ddd1d5b29c605ba79f21cc')
- username: "hita123"
- password: "\$2b\$10\$lnvmm38SXP6QMMfNUIsvB.x0KJQ3.KjzTlKdpXQfTsubwzc4n59nq"
- role: "customer"
- securityQuestions: Array (3)
 - 0: Object
 - 1: Object
 - 2: Object
- address: Object
- skills: Array (empty)
- rating: 0
- points: 0
- bookings: Array (empty)

Backend Deployment

→ ↺ dashboard.render.com/web/srv-d3998tjuibrs73abti4g/events

📄 ☆ 🗂️ 🔍 ⬇️ 👤

M My Workspace ⌵

Dashboard

hand2hand-backend

Events

Settings

MONITOR

Logs

Metrics

MANAGE

Environment

Shell ⚡

Scaling ⚡

Previews

Disk ⚡

Changelog ⌵

Invite a friend

Contact support

Render Status

🌐 hand2hand-backend ⌵

🔍 Search ^ K

+ New ⚡ Upgrade ? 👤

🌐 WEB SERVICE

hand2hand-backend

Node Free Upgrade your instance →

Connect ⌵ Manual Deploy ⌵

Service ID: srv-d3998tjuibrs73abti4g 📄

🔄 Hema-Narala / hand2hand-backend ⚙️ main

https://hand2hand-backend.onrender.com 📄

📘 Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more. Upgrade now

🔍 Filter events 31 ⌵

Service recovered

✅ Recently failed instances are now reporting healthy.

April 14, 2026 at 8:12 PM

Instance failed: 2ftsrr

⚠️ Exited with status 1 while running your code. Check your [service logs](#) for more information.

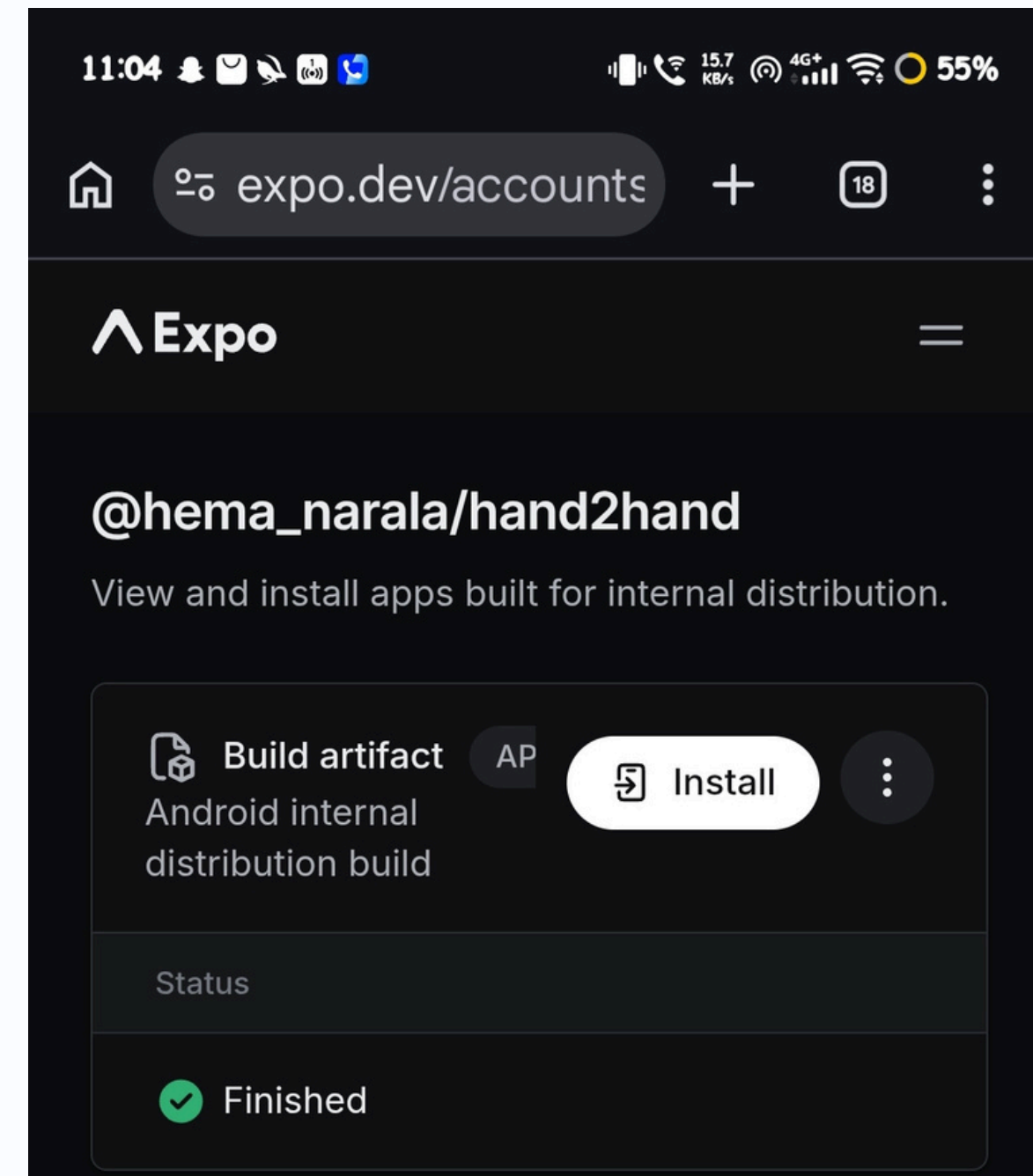
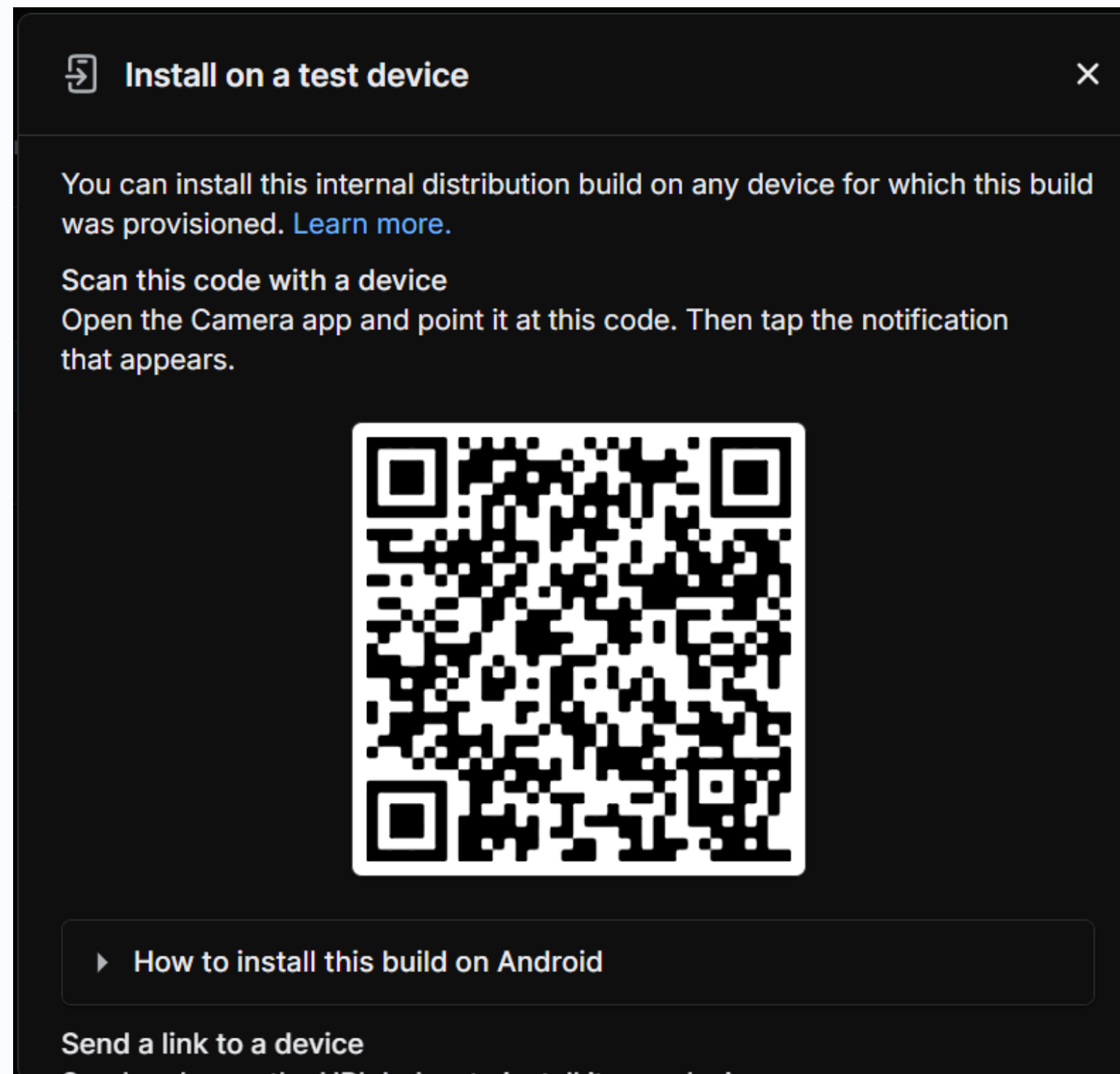
April 14, 2026 at 8:12 PM

Debug ⌵

Service recovered

✅ Recently failed instances are now reporting healthy.

Application Link



Link: <https://shorturl.at/Y47o7>

Conclusion

TheHand2Hand project connects customers with local laborers through a simple and reliable digital platform built using the MERN stack (MongoDB, Express.js, React.js, Node.js). It provides key features such as user registration, worker profiles portfolio building , job posting , Booking system, chat system within the application, ratings & reviews , search and filters and service management through an interactive React-based interface. The backend efficiently handles data using Node.js, Express, and MongoDB Atlas, with JWT-based authentication ensuring security. The platform is deployed online with the backend on Render and created an APK file for frontend, providing a scalable cloud-based solution. With additional improvements and future enhancements, Hand2Hand aims to further simplify service access, support local workers, and promote community development through technology.

References

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The background features a white central area framed by a light gray border. Four dark blue geometric shapes, resembling stylized corners or petals, are positioned at the top-left, top-right, bottom-left, and bottom-right. The text "Thank You" is centered in a bold, dark blue font.

**Thank
You**